IMPERIAL

Electricity Markets and the Power Grid

Cristina Vera and Dr Esma Koca

Agenda

1 **Grid Architecture**

2 **Electricity Market Fundamentals**

3 **Electricity Market Models & the UK Markets**

4 **Electricity Pricing**

5 **Electricity Analytics**

6 **Data Explorer**

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1 Grid Architecture

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6 Data Explorer

1. Grid Architecture

Backbone system of modern society, delivering **electricity** to **homes, industries and services**.

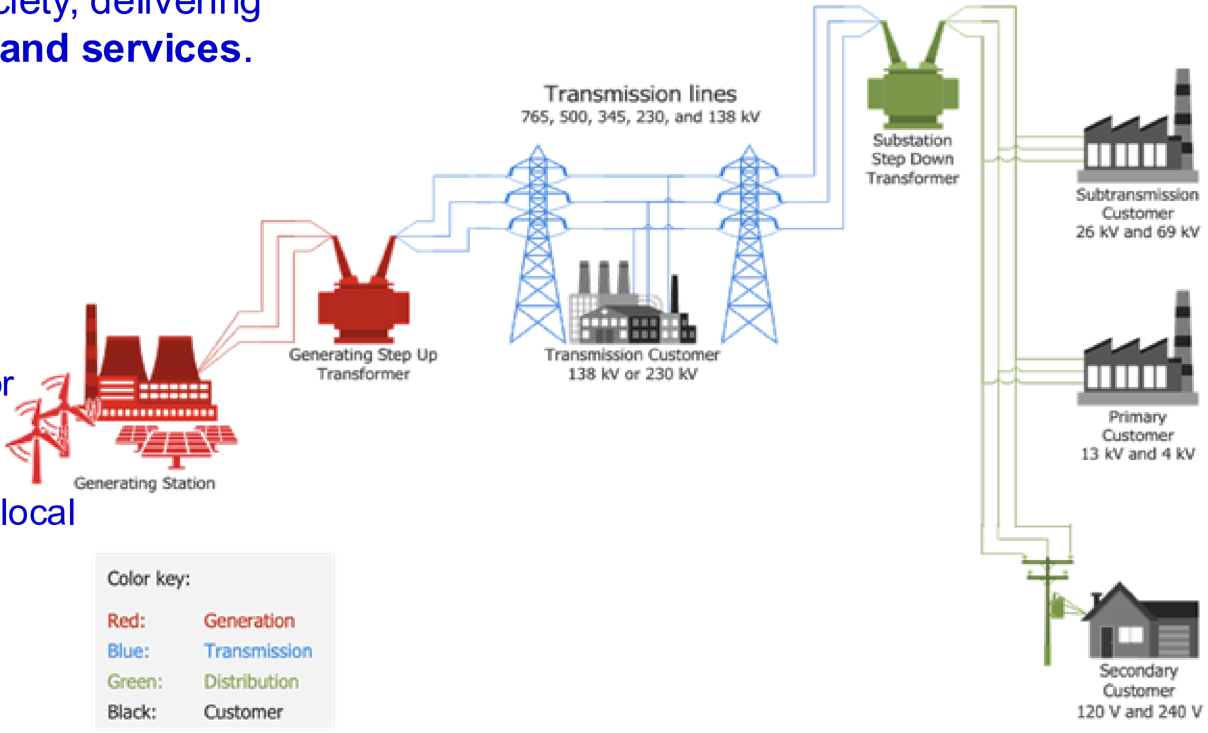
The main components:

Generation – where electricity is produced.

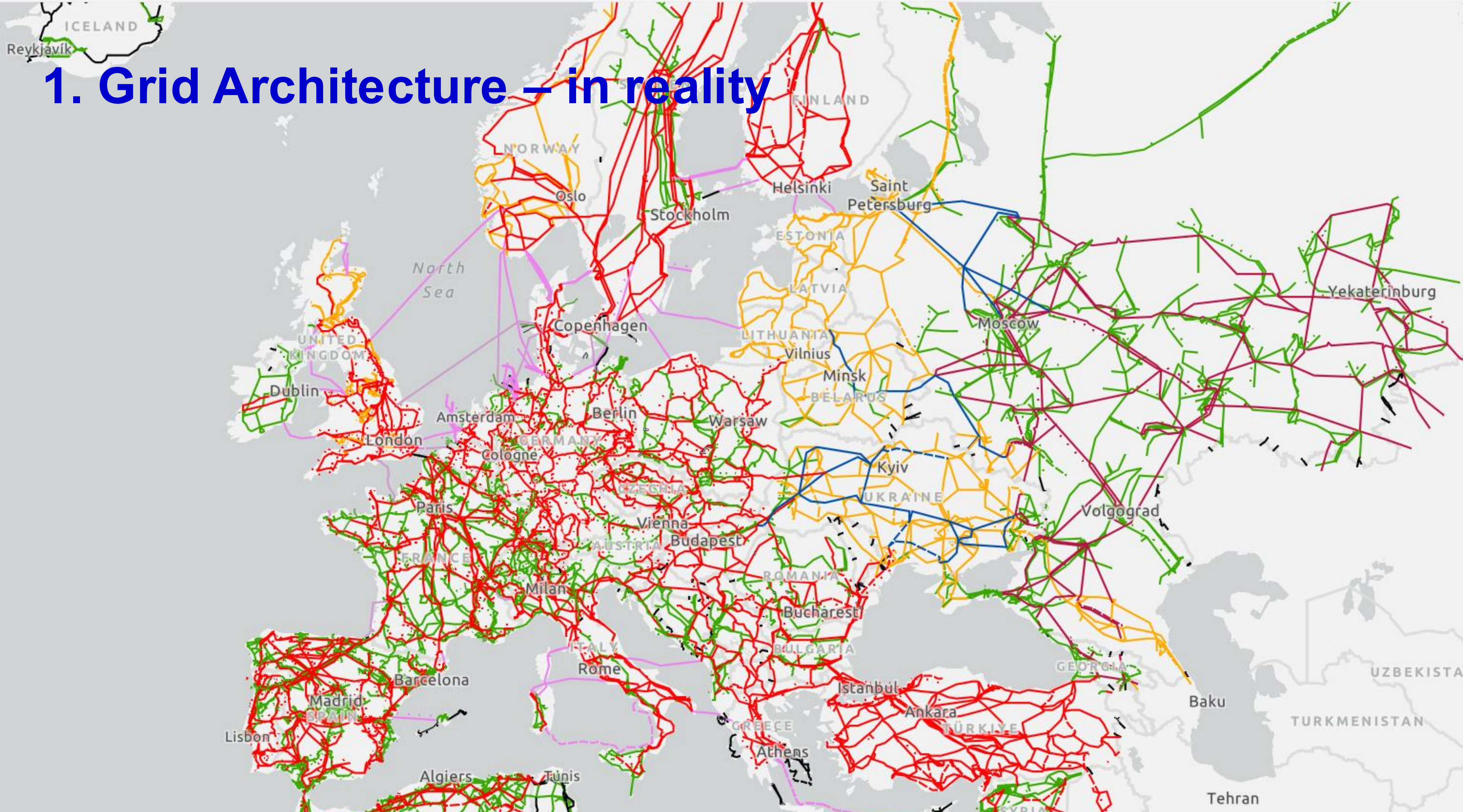
Transmission – high-voltage lines for long distance.

Distribution – lower-voltage lines to local consumers.

Consumer – the demand of the electricity.



1. Grid Architecture – in reality



ENTSO-E Grid Map

Often called the most complex machine ever built and it’s only getting more complex.

1. Grid Architecture – the Modern Grid

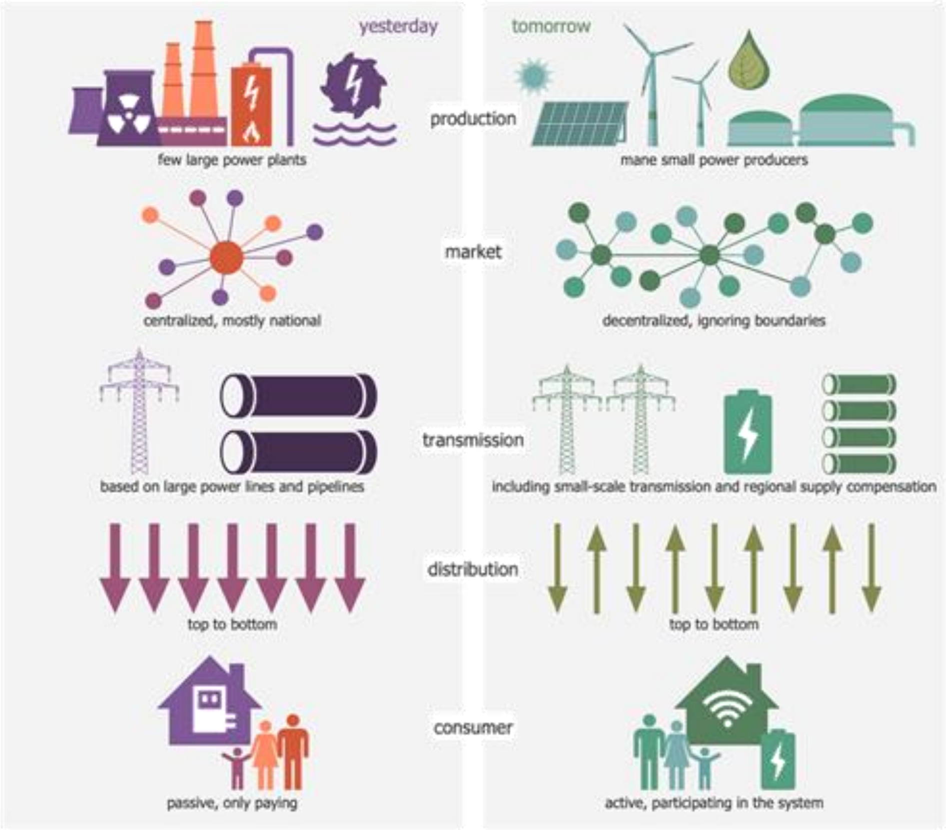
Yesterday's Grid:

- Few large centralized power plants.
- Reliance on big transmission lines.
- One-way top-down distribution.
- Passive consumers.

Tomorrow's Grid:

- Many small renewable and variable producers.
- Mix of large & small transmission with regional balancing.
- Two-way distribution.
- Active consumers (prosumers).

The power grid's increasing complexity requires **smart technologies**.



Source: Energy Industry Infographics Solution | ConceptDraw.com

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2 **Electricity Market Fundamentals**

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2. Electricity Market Fundamentals

Synchronous Balance

Diversity of Generation

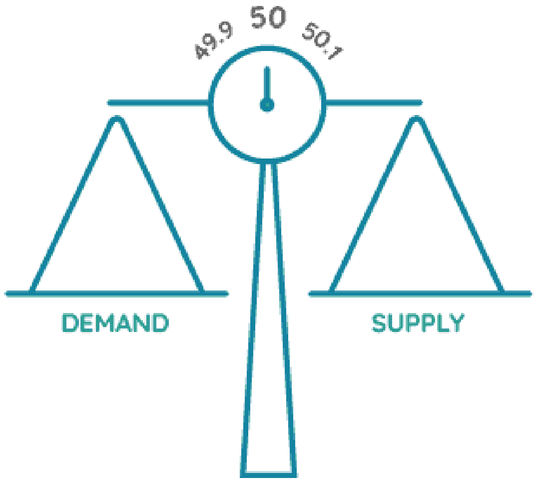
Volatile Prices

Natural Monopoly

2. Electricity Market Fundamentals

Synchronous Balance

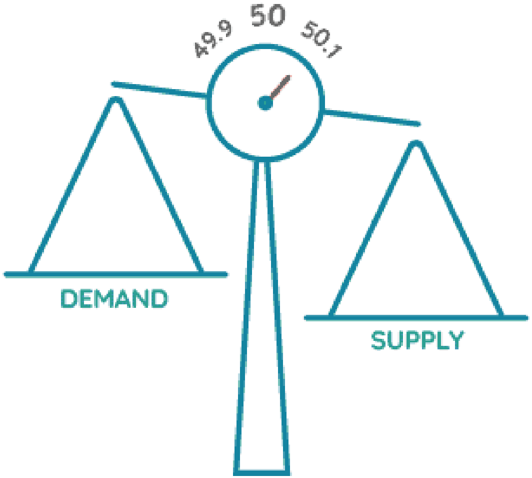
Supply = Demand + Network Losses
across the entire grid at ~50 Hz



Source: https://sympower.net/articles/the-differences-in-demand-response-balancing-services-explained

2. Electricity Market Fundamentals

Synchronous Balance

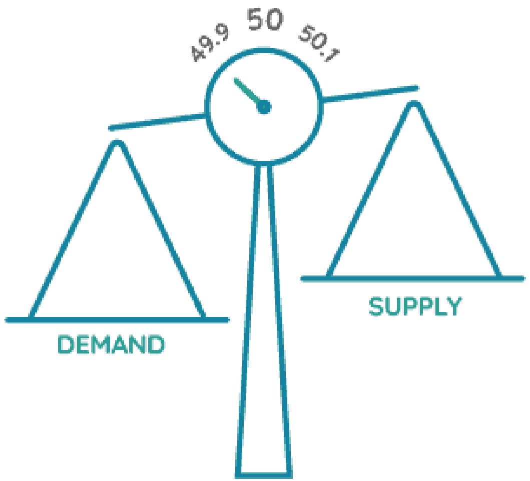


If **supply > demand** → Frequency rises → risk of equipment damage, protective trips, and instability.

Source: https://sympower.net/articles/the-differences-in-demand-response-balancing-services-explained

2. Electricity Market Fundamentals

Synchronous Balance



If **demand > supply** → Frequency drops → risk of brownouts, equipment stress, or blackouts.

Below 49.5 Hz triggers automatic demand disconnection. Operators have ~10 seconds to react.

Source: https://sympower.net/articles/the-differences-in-demand-response-balancing-services-explained

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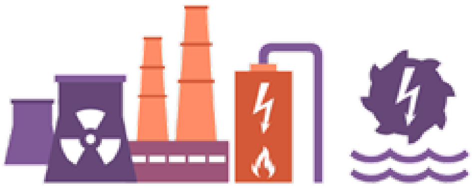
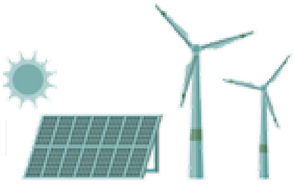
2. Electricity Market Fundamentals

Diversity of Generation

dispatchable sources → variable renewables



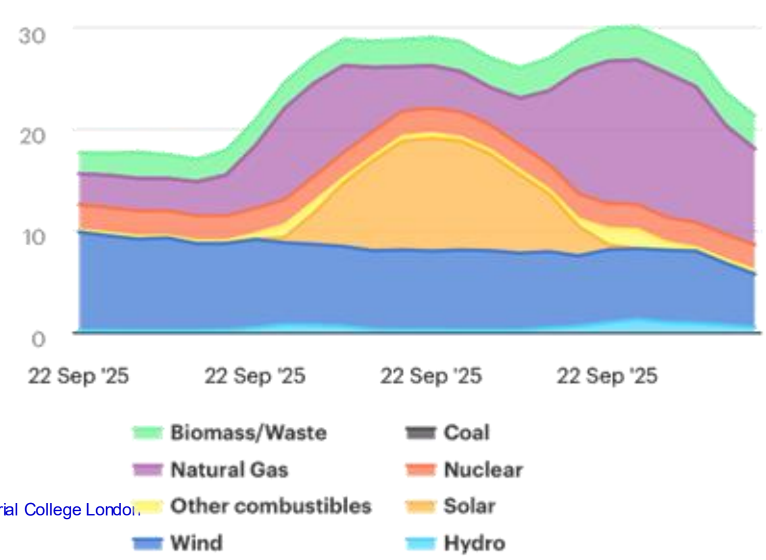


2. Electricity Market Fundamentals

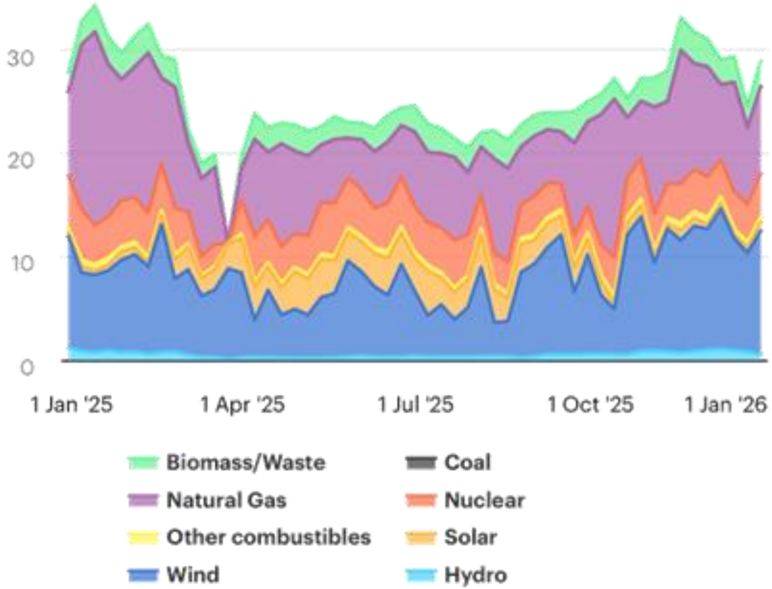
Diversity of Generation

Unlike dispatchable generation, variable renewables such as solar and wind output fluctuates with the time of day and the season.

Daily electricity generation in the UK (GW)



Annual electricity generation in the UK (GW)

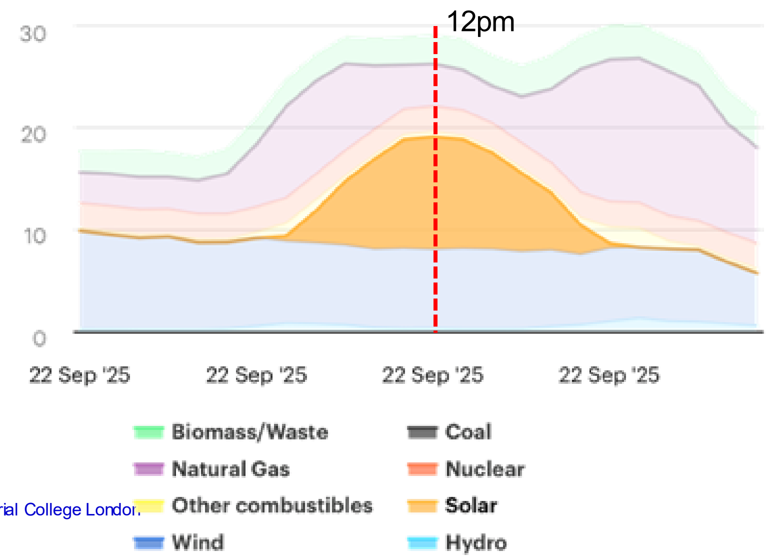


2. Electricity Market Fundamentals

Diversity of Generation

Solar generation is limited to daylight hours and is greatest during summer, when days are longest.

Daily electricity generation in the UK (GW)



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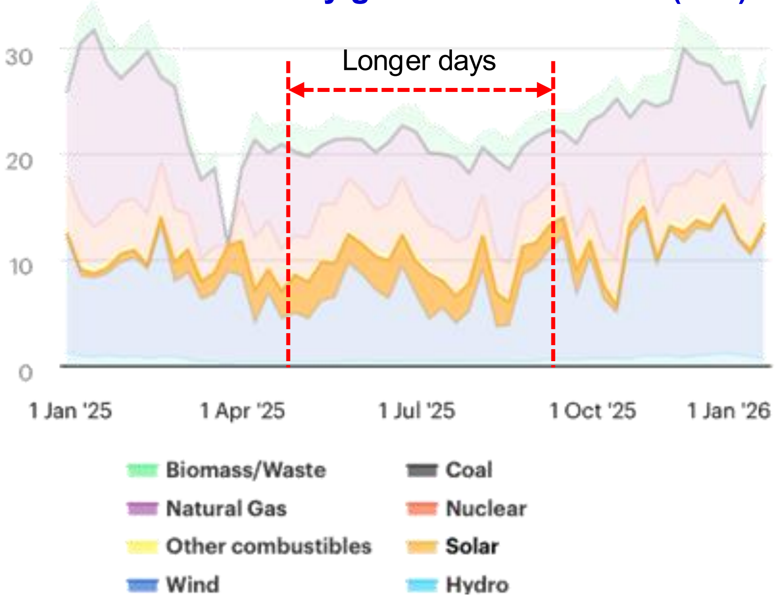
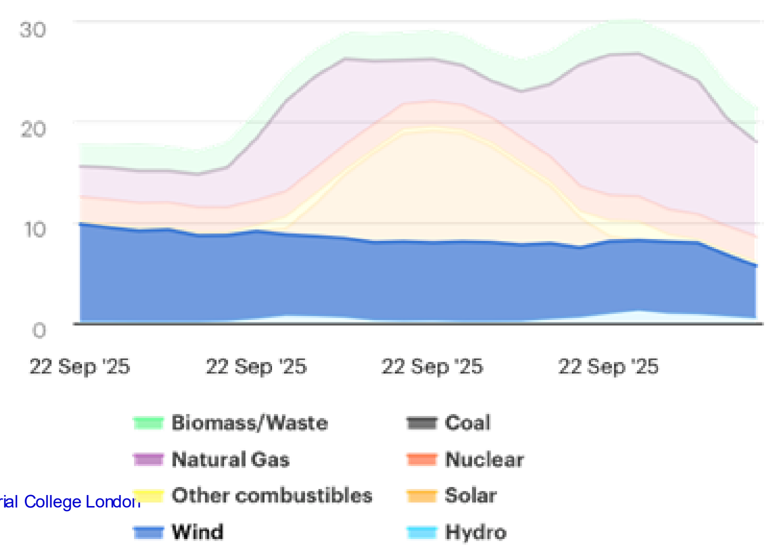


2. Electricity Market Fundamentals

Diversity of Generation

Wind generation is available at any hour but varies with wind speed, and is generally higher during winter months.

Daily electricity generation in the UK (GW)



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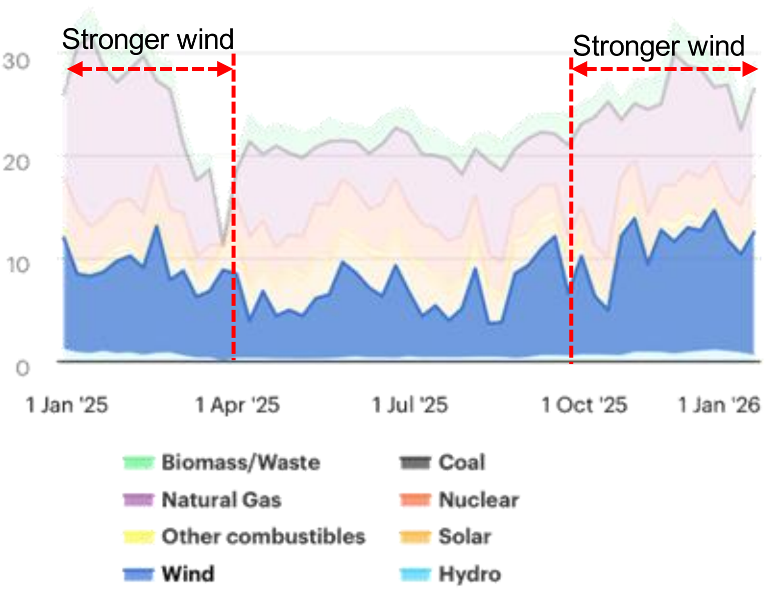
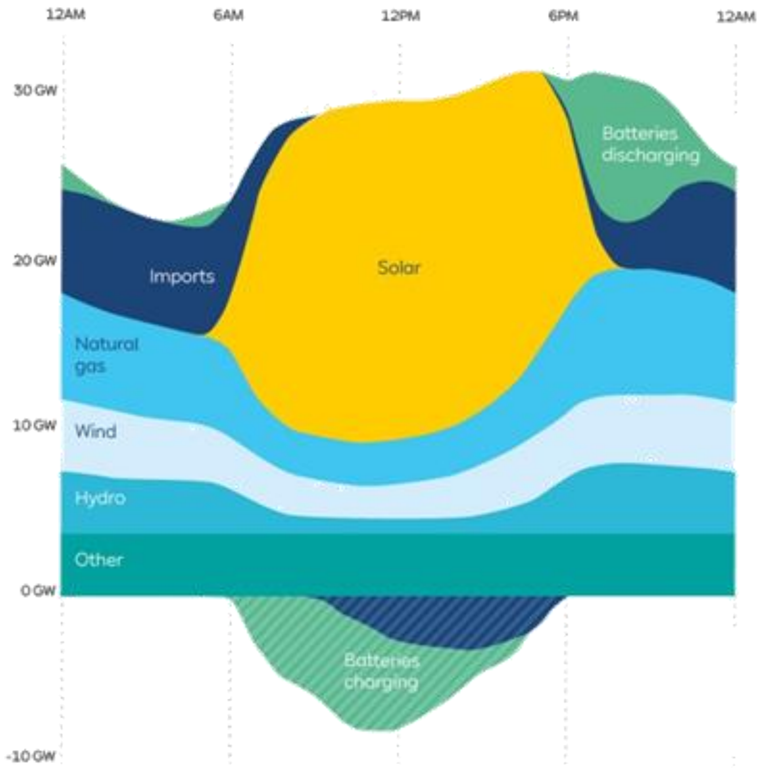


2. Electricity Market Fundamentals

Diversity of Generation

Renewable variability can be compensated by batteries and grid interconnection.



 Source: California Independent System Operator

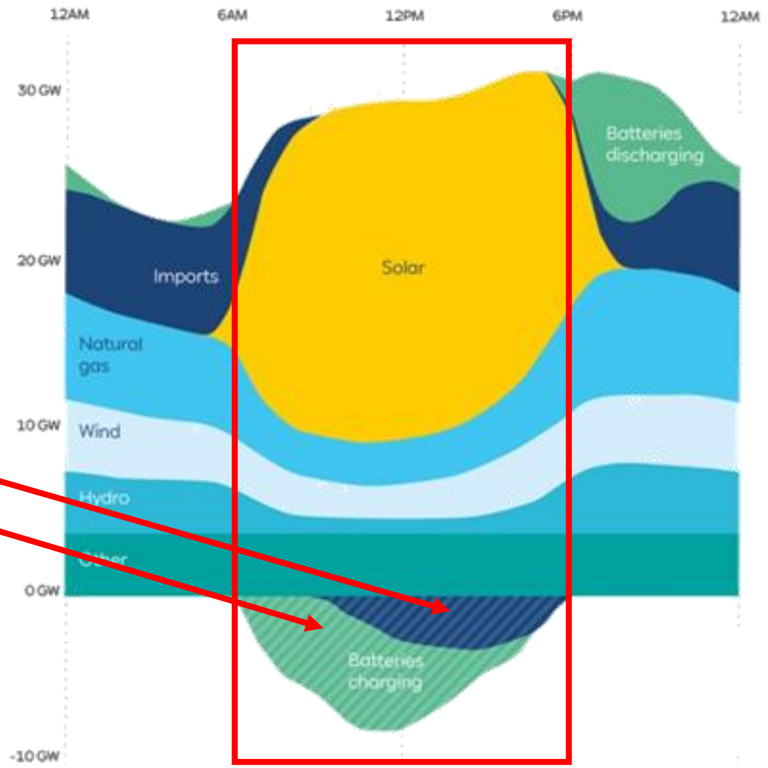
2. Electricity Market Fundamentals

Diversity of Generation

Renewable variability can be compensated by batteries and grid interconnection.

Excess generation:

- Export energy
- Charge batteries



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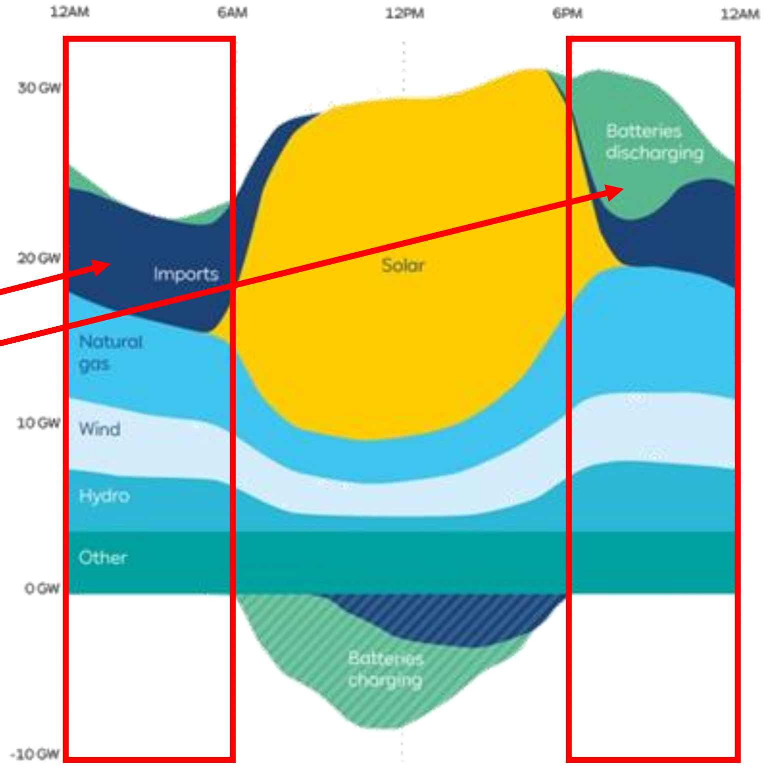
Insufficient generation:
- Import energy
- Discharge batteries



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Diversity of Generation

Renewable variability can be compensated by batteries and grid interconnection.

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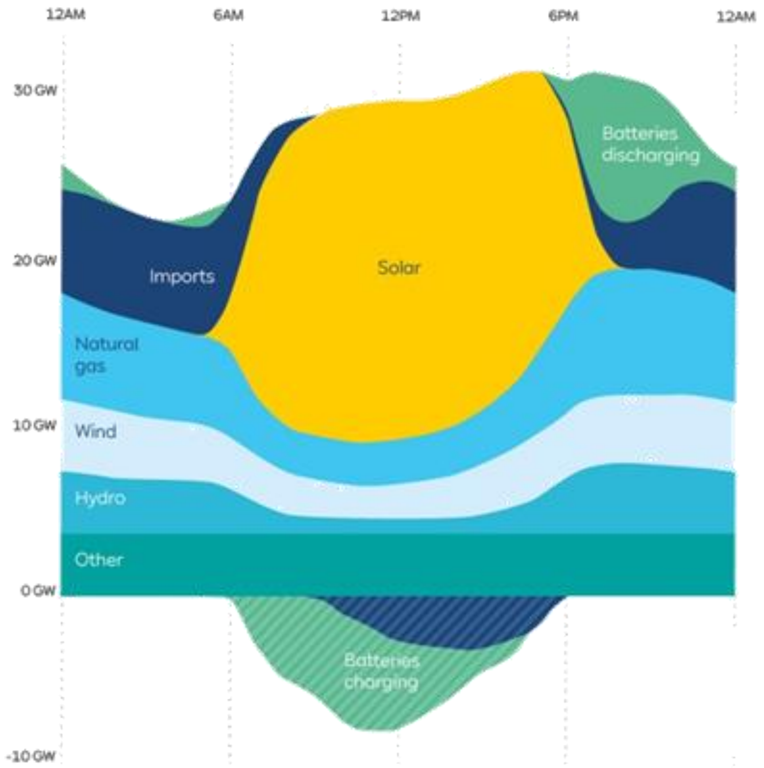
Electricity cannot be economically stored at scale (yet) for long periods of time.



 Source: California Independent System Operator

2. Electricity Market Fundamentals

Synchronous Balance

Diversity of Generation

Volatile Prices

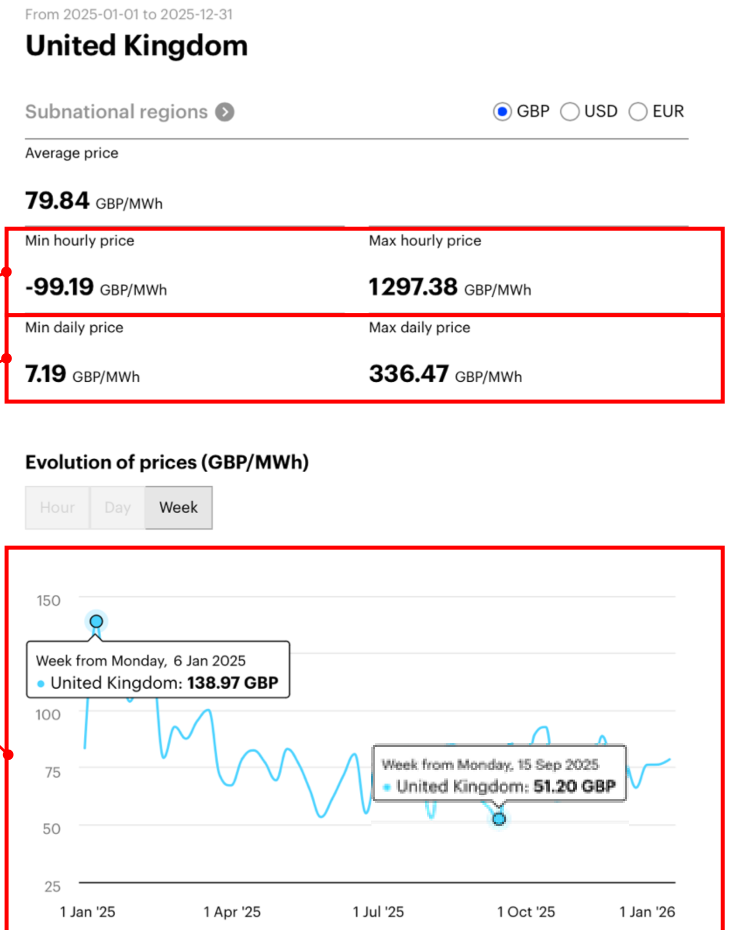
Natural Monopoly

2. Electricity Market Fundamentals

Volatile Prices

UK **spot prices** in 2025:

- Hourly absolutes from -99.19 £/MWh to 1297.38 £/MWh
  → 1408% swing increase
- Daily averages from 7.19 £/MWh to 336.47 £/MWh
  → 4580% increase
- Weekly averages from 51.20 £/MWh to 138.97 £/MWh
  → 171% increase



Source: International Energy Agency

2. Electricity Market Fundamentals

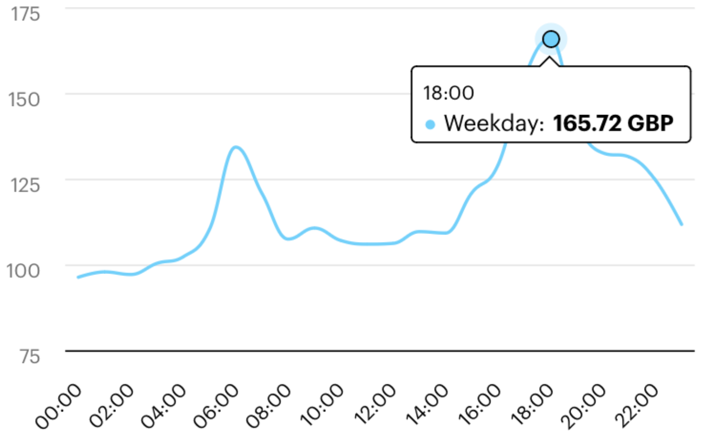
Volatile Prices

What explains Spot Price volatility?

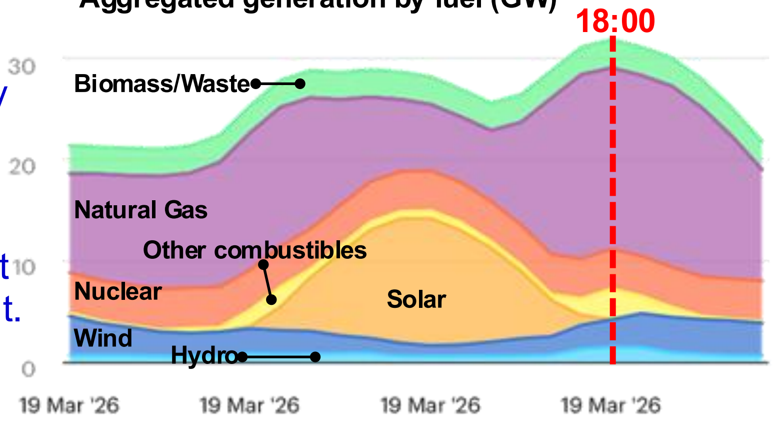
Spot price is the real-time price of keeping supply and demand of electricity matching at every moment.

- **Demand spikes or supply failures:** sudden imbalance hits prices almost instantly
- **Negative prices:**
    - Grid is oversupplied, generators pay to offload electricity
    - Surplus of renewable + Inflexible generation: Some plants can't ramp down quickly
- **Price spikes:**
    - **Energy mix & merit order:** spot price is set by the most expensive plant needed to meet demand at that moment.

Hourly prices (GBP/MWh)



Aggregated generation by fuel (GW)



Source: International Energy Agency

2. Electricity Market Fundamentals

Volatile Prices

Price volatility does not hit immediately the electricity consumer:

$$Retail\ Price = \underbrace{Wholesale\ Price}_{\textbf{HIGHER VOLATILITY}} + \underbrace{Power\ Grid\ Cost + taxes\ \&\ levies}_{\textbf{FIX COSTS}}$$

$$\overbrace{\underbrace{forwards\ \&\ futures}_{\substack{\textbf{PRE-AGREED PRICE}\\ \textbf{FOR FUTURE VOLUME}\\ \textbf{Reduced uncertainty}}} + \underbrace{spot\ prices}_{\substack{\textbf{REAL-TIME}\\ \textbf{PRICE}}}}$$

- Electricity suppliers hedge: forwards, futures, and spot purchases
- The higher the share of forwards/futures → the longer the lag before volatility reaches consumers
- The higher the share of spot purchases → the faster wholesale volatility transmits to retail

Source: International Energy Agency

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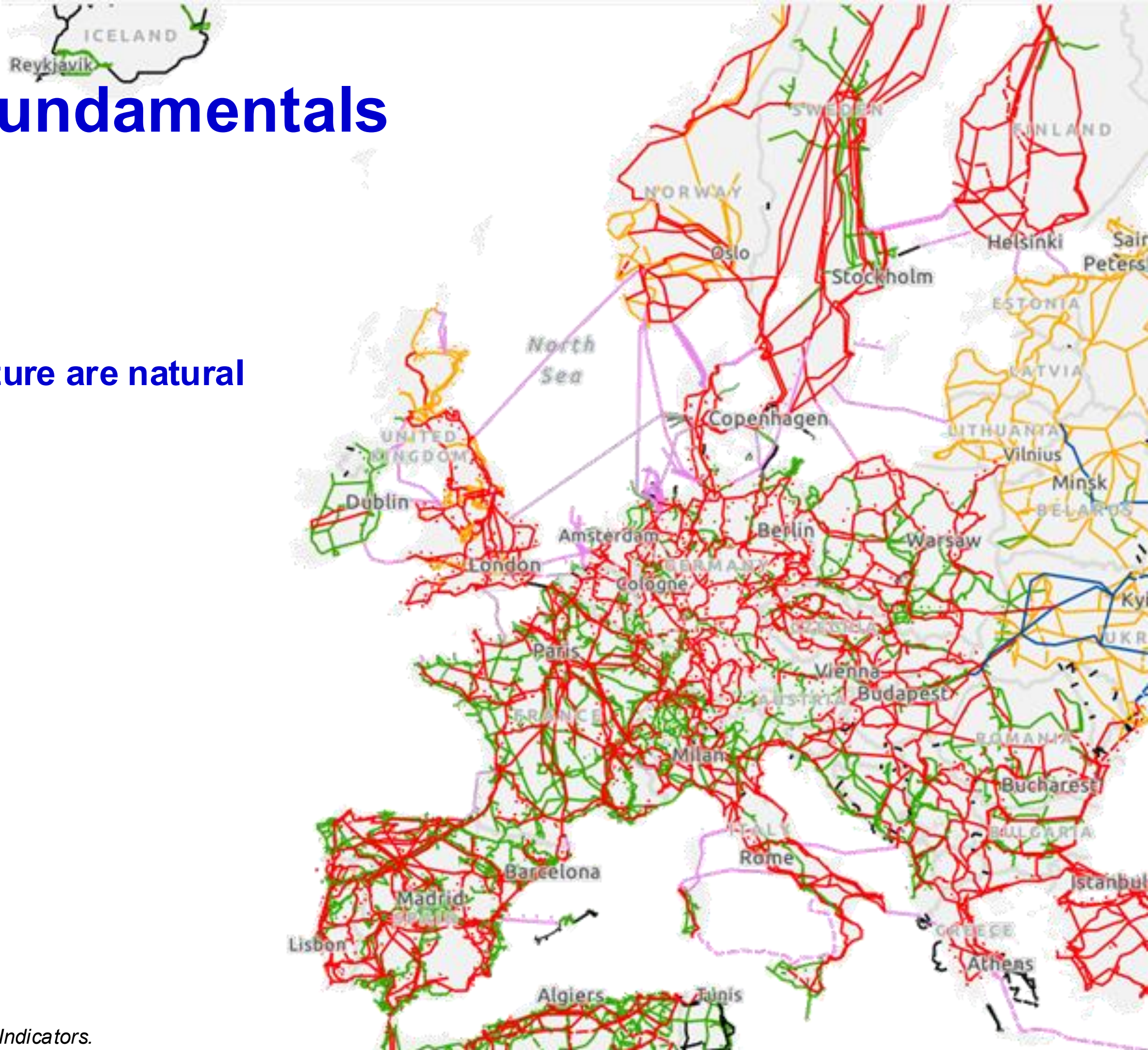
2. Electricity Market Fundamentals

Natural Monopoly

Transmission and distribution infrastructure are natural monopolies → it is build once.



Imperial College London

Source: PwC & ACER (2023) *Unit Investment Cost Indicators.*

2. Electricity Market Fundamentals

Natural Monopoly

Transmission and distribution infrastructure are natural monopolies → it is build once.

- **Extremely high fixed costs**: overhead lines up to £4M/mile; underground cables up to £6.8M/mile.
- **Duplication is economically irrational**: double capital costs with no efficiency gain
- **Planning & permitting barriers**: years of environmental and regulatory approvals; land acquisition is legally complex

Single infrastructure, shared use: all generators and suppliers access the same infrastructure, regulated to ensure fair access

Source: PwC & ACER (2023) *Unit Investment Cost Indicators.*

2. Electricity Market Fundamentals

Natural Monopoly

Eastern Green Link 3 (EGL3)

- 525 kV HVDC subsea cable linking Scotland and England
- **~422 miles**, onshore and offshore sections
- 2 GW (enough for up to 2 million homes)
- Joint venture between **National Grid Electricity Transmission** and **SSEN Transmission**
- **High CAPEX and long projects: ~£3bn** worth of contracts; starts in 2028 to end of 2033 (~6 years)

Source: https://eandt.theiet.org/2026/03/04/key-milestone-reached-delivery-uks-largest-electricity-transmission-project
https://www.hitachienergy.com/news-and-events/customer-stories/eastern-green-link-3

2. Electricity Market Fundamentals

Natural Monopoly

Single infrastructure; entrusted to companies to run it.

Critical infrastructure; direct state ownership or strict regulatory oversight.

Model	**Examples**
State owned	France (RTE), Germany (partially), most of Eastern Europe
Regulated private	UK (National Grid), Italy (Terna), Spain (REE)

Source: Entsoe-eu

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3. Electricity Market Models

Liberalisation

A. Vertically Integrated Monopoly

B. Single Buyer Model

C. Wholesale Competition

D. Wholesale & Retail Competition

Reference: Cretì, A. and Fontini, F. (2019) *Economics of electricity: markets, competition and rules*. Chapter 4: The Four Market Designs of the Electricity System. Cambridge: Cambridge University Press. Available at: https://doi.org/10.1017/9781316884614

3. Electricity Market Models

Liberalisation

A. Vertically Integrated Monopoly

B. Single Buyer Model

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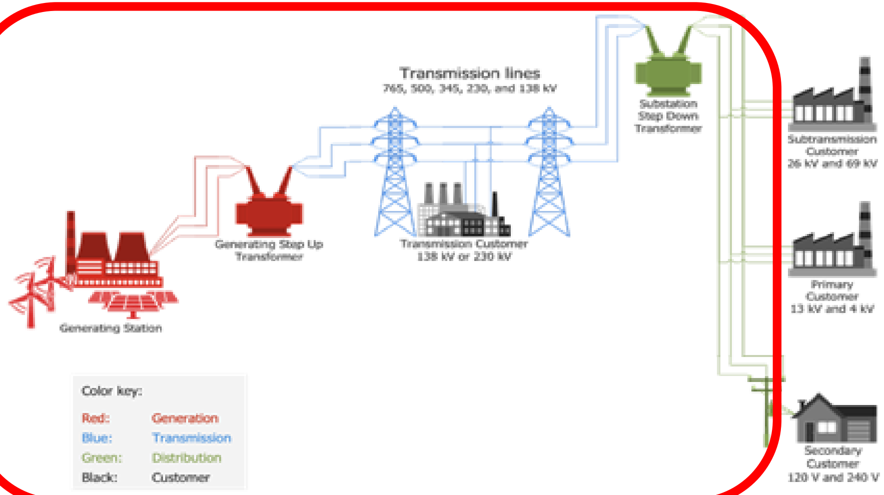
D. Wholesale & Retail Competition

- Single state-owned utility controls generation, transmission & retail.
- Prices are regulated (often subsidised) — no market price signal.



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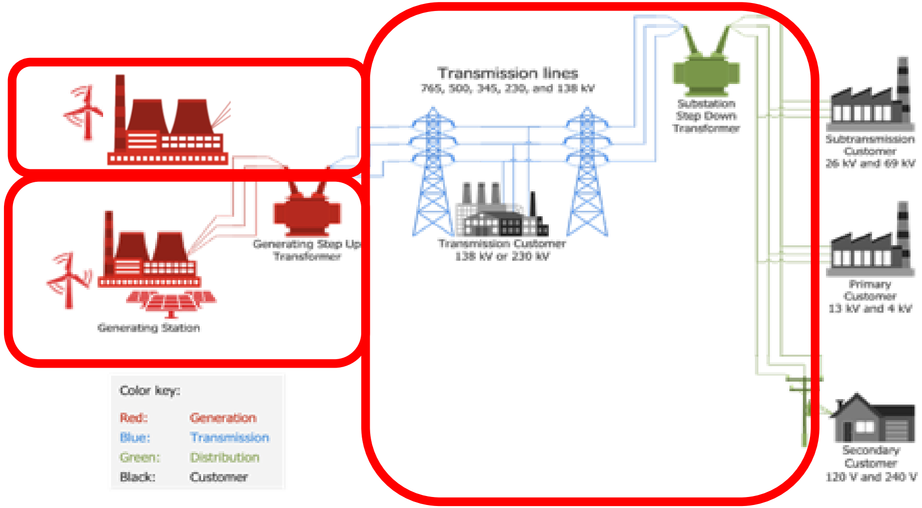
3. Electricity Market Models

Liberalisation

- A. Vertically Integrated Monopoly
- **B. Single Buyer Model**
- C. Wholesale Competition
- D. Wholesale & Retail Competition

- Independent generators sell to one monopsonist buyer – usually first step towards market opening.
- Buyer sells to end consumers at regulated tariffs.



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3. Electricity Market Models

Liberalisation

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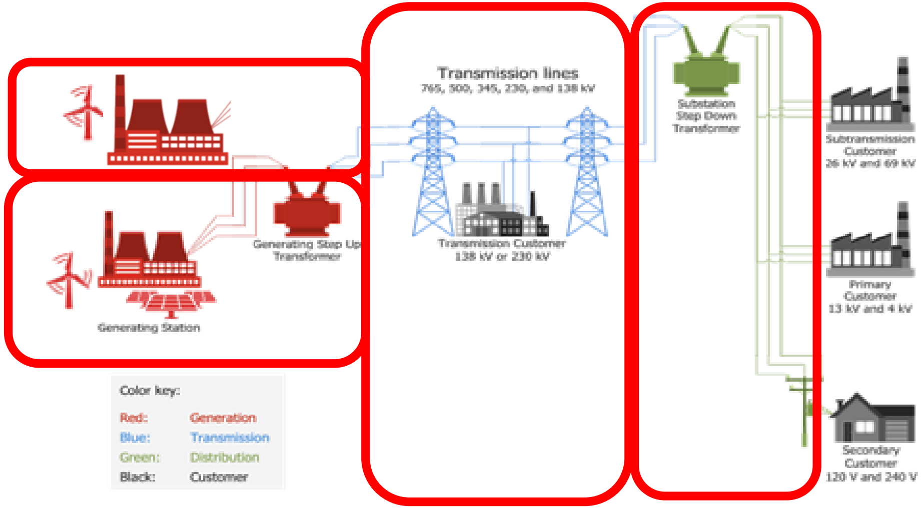
D. Wholesale & Retail Competition

- Competition exists only at the generation level.
- Transmission and distribution remain regulated monopolies & end consumers are served by a regulated retailer.
- Market prices emerge from bidding.



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3. Electricity Market Models

Liberalisation

- A. Vertically Integrated Monopoly
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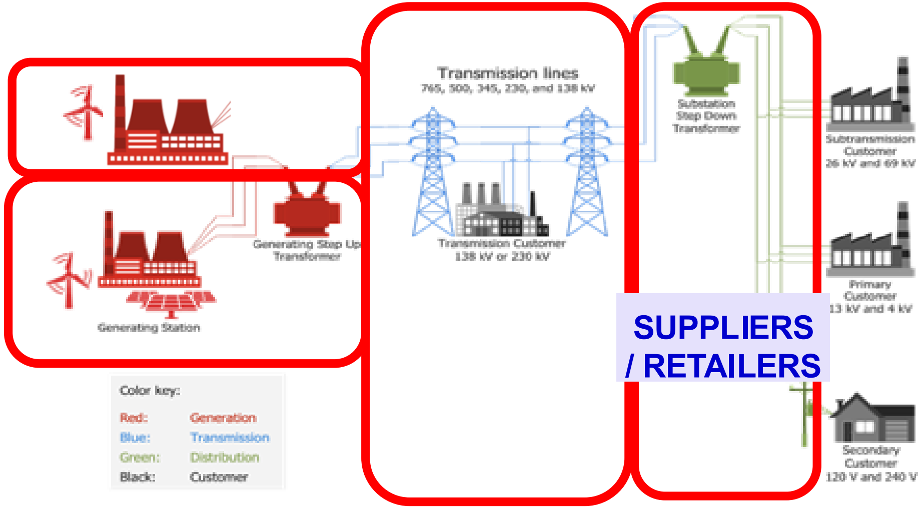
- **D. Wholesale & Retail Competition**

- Suppliers compete on price/products.
- The network (wires) remains regulated, but the commercial relationship with the end customer is liberalised.



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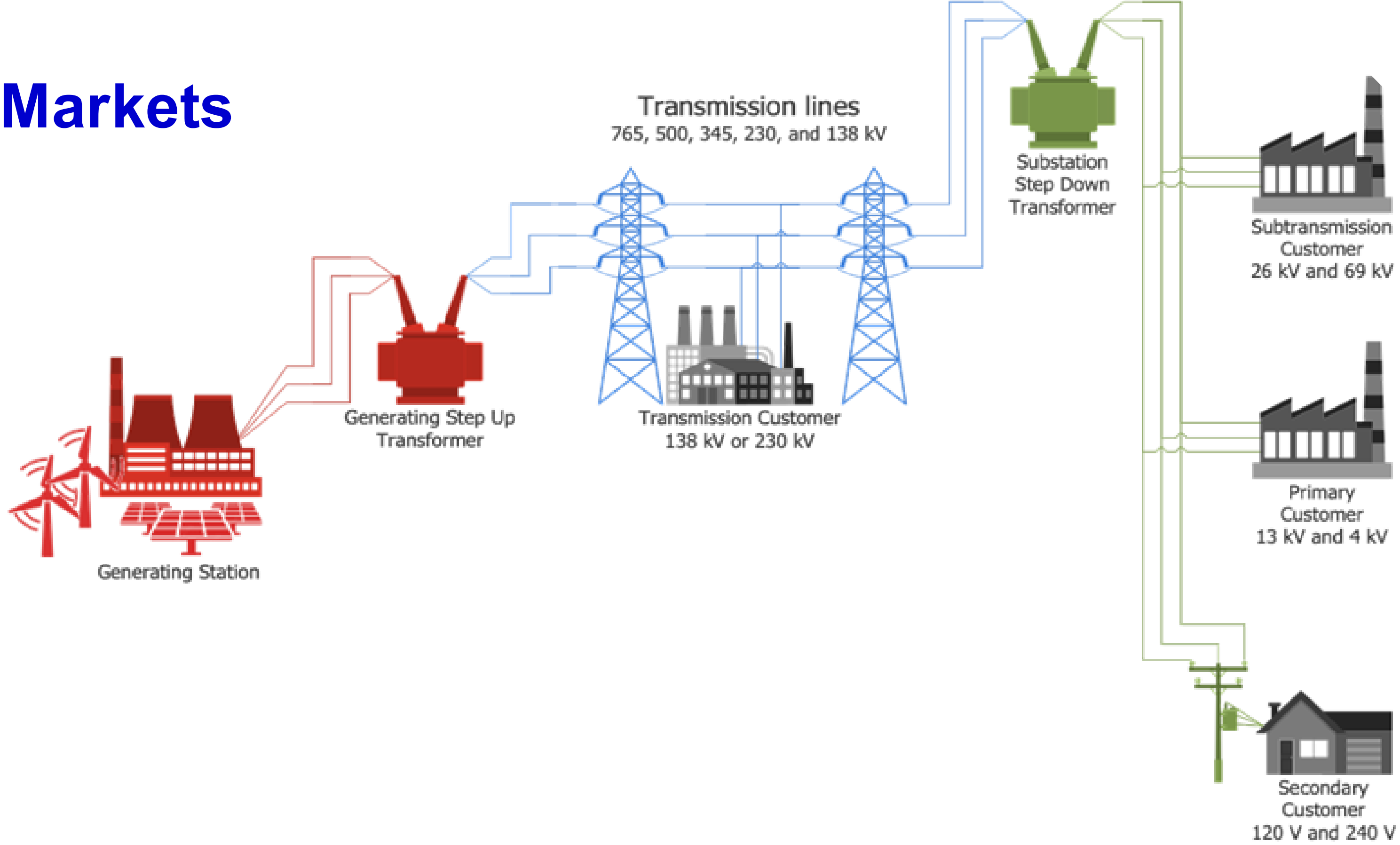
3. UK Markets

Market	Who	When	What	How
Capacity market	Generators	1-4 Years	**Total generation capacity available; ”insurance policy”**	Generators compete at auction for capacity contracts. Winners are paid to guarantee capacity (4h notice), even if not dispatched
Wholesale market	Generators & Retailers	Years to hours	**When and how much electricity output and at what price**	Merit order buy/sell electricity via forwards, futures, and day-ahead auctions
Gate closure (milestone): 1 hour before delivery, commercial trading stops → results of the wholesale market are transferred to the System Operator to manage the imbalances.				
Balancing mechanism	System Operator	Real-time	**Real-time supply-demand balancing**	Generators are paid to increase or reduce their previously agreed generation
Ancillary services	System Operator	Real-time	**Granular adjustments to the system**	Frequency response, reserve, reactive power, and black start

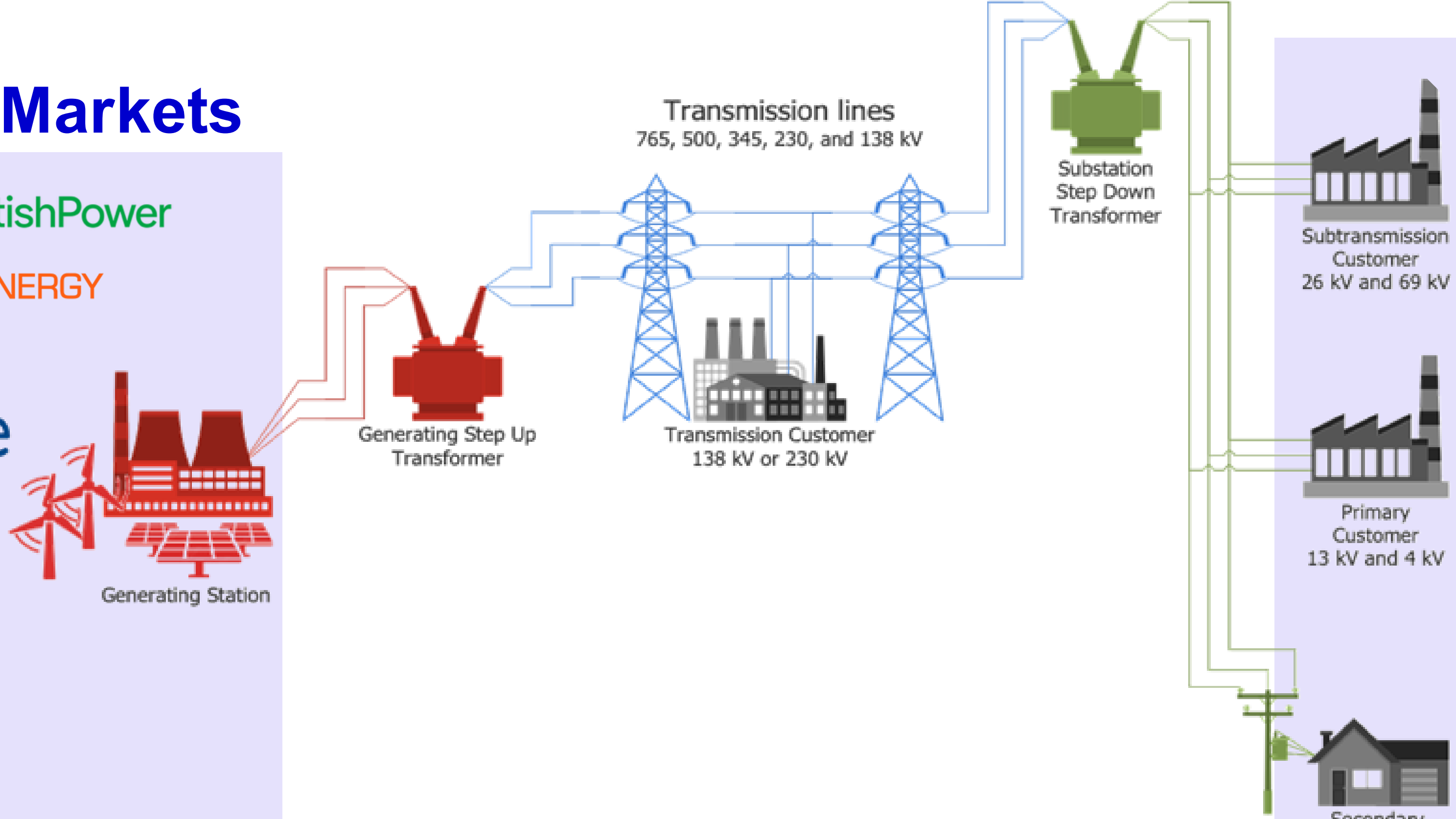
Reference: Modo Energy - Electricity Markets in Great Britain - The Energy Academy (Series Three)

3. UK Markets



Transmission lines
765, 500, 345, 230, and 138 kV
Substation
Step Down
Transformer
Subtransmission
Customer
26 kV and 69 kV
Generating Step Up
Transformer
Transmission Customer
138 kV or 230 kV
Primary
Customer
13 kV and 4 kV
Generating Station
Secondary
Customer
120 V and 240 V

3. UK Markets



Produce electricity from various sources

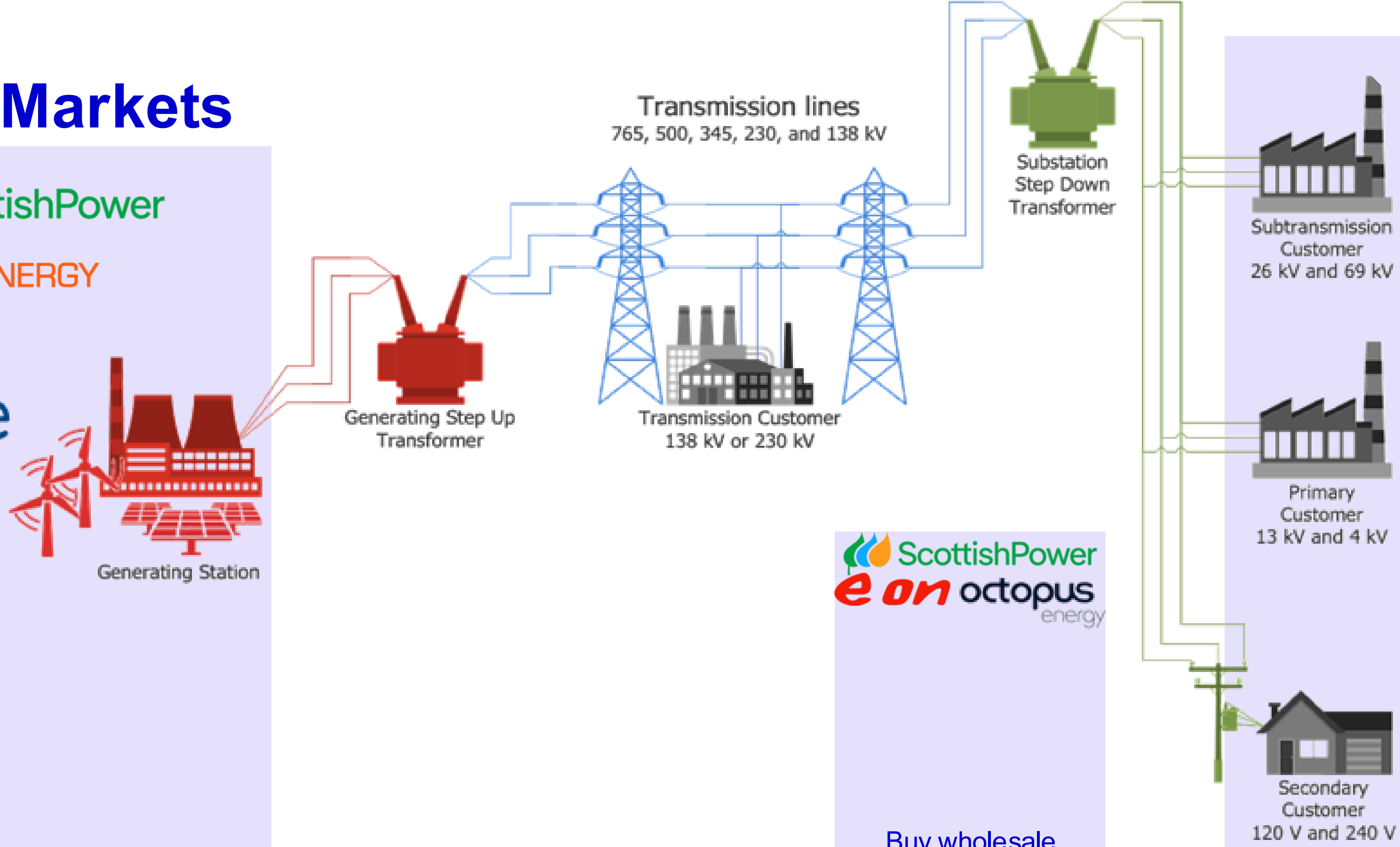
GENERATORS

Buy retail electricity

END-USERS

3. UK Markets

ScottishPower
EDF ENERGY
RWE
sse



Produce electricity from various sources

GENERATORS

ScottishPower
e on
octopus energy

Buy wholesale electricity to sell to end-users

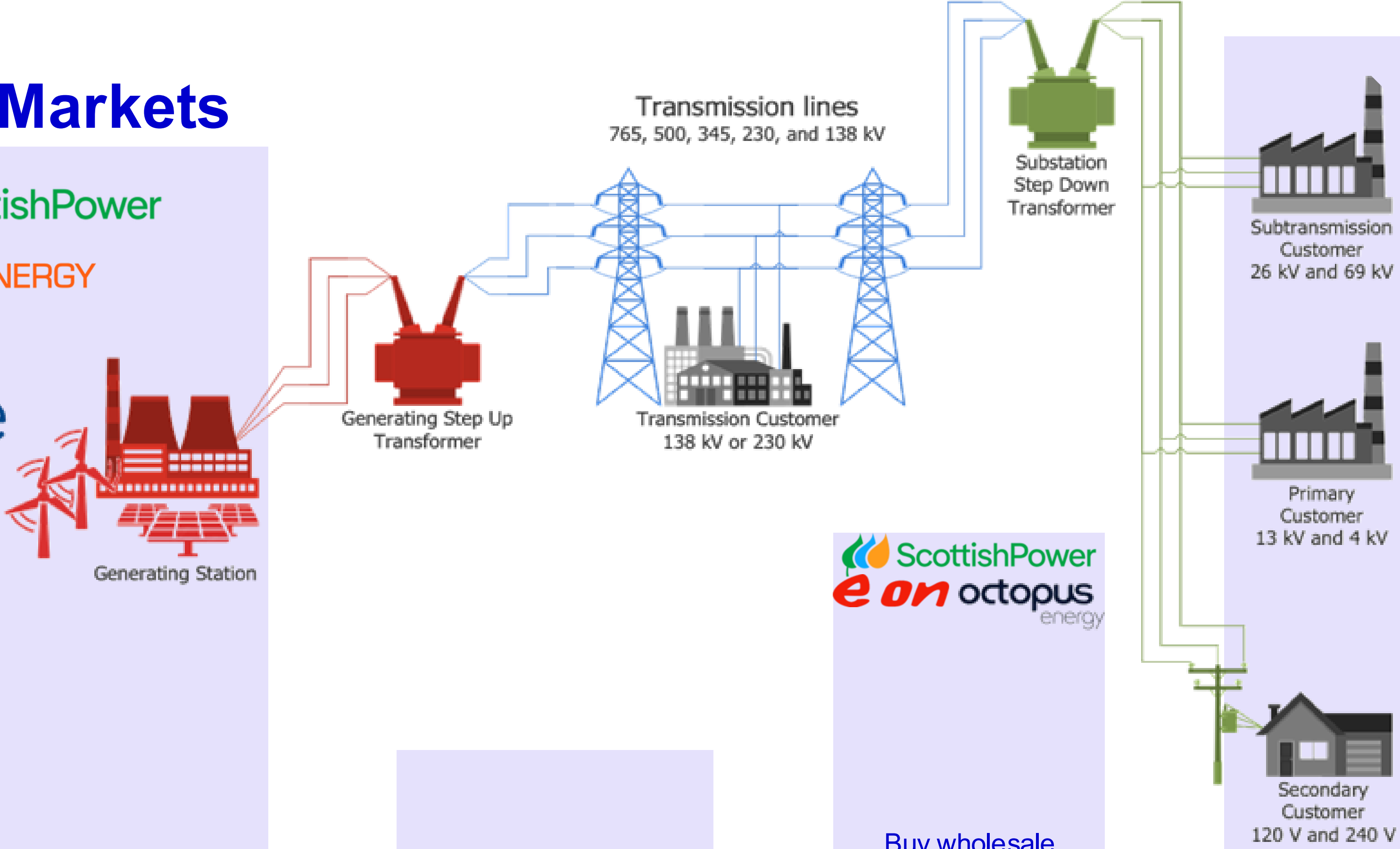
SUPPLIERS / RETAILERS

Sell electricity

Buy retail electricity

END-USERS

3. UK Markets



ScottishPower
EDF ENERGY
RWE
sse

Produce electricity from various sources

GENERATORS

Sell electricity

Buy/Sell electricity to meet demand

WHOLESALE MARKET

Buy electricity

ScottishPower
e on
octopus energy

Buy wholesale electricity to sell to end-users

SUPPLIERS / RETAILERS

Sell electricity

Buy retail electricity

END-USERS

3. UK Markets

ScottishPower
EDF ENERGY
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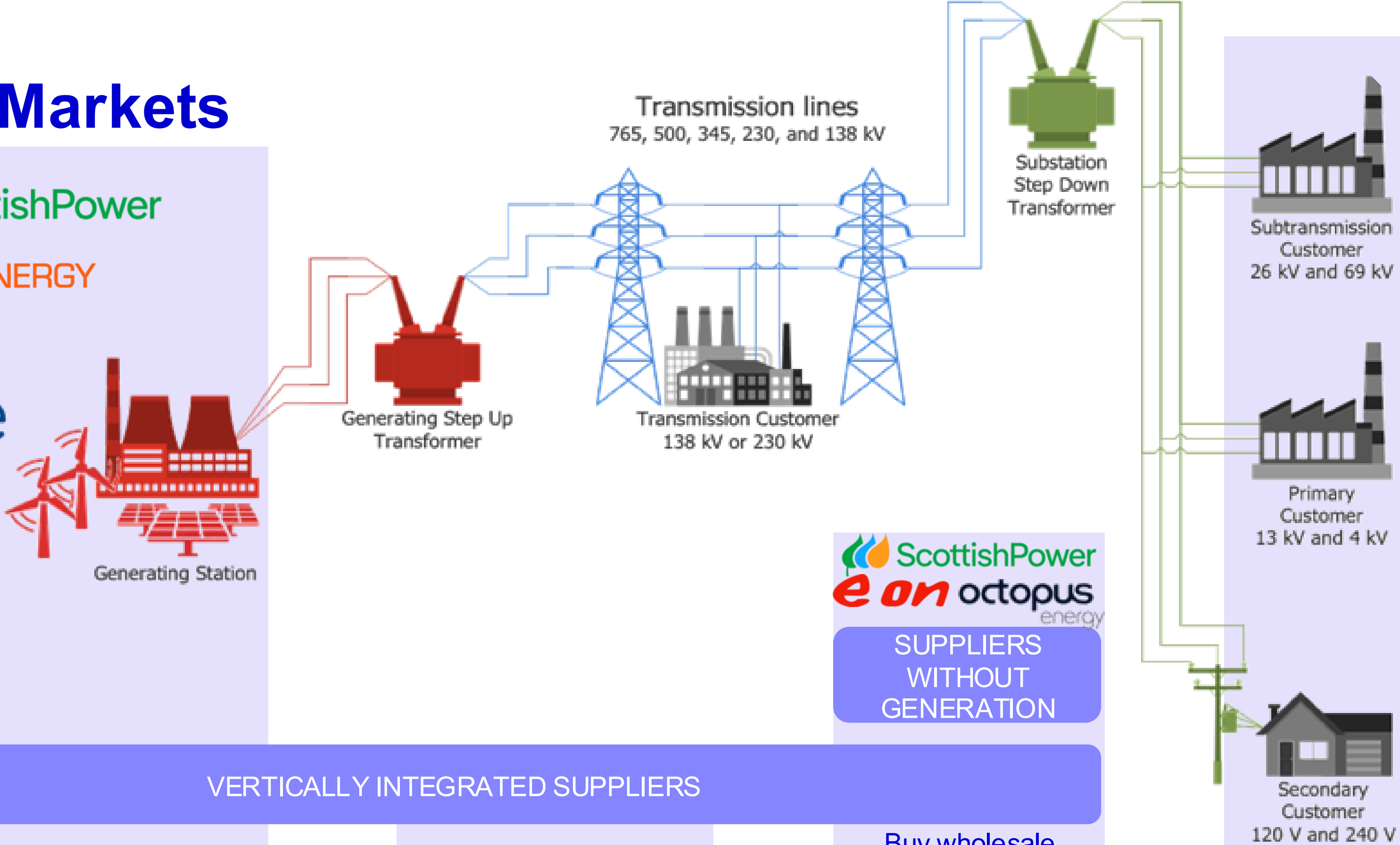


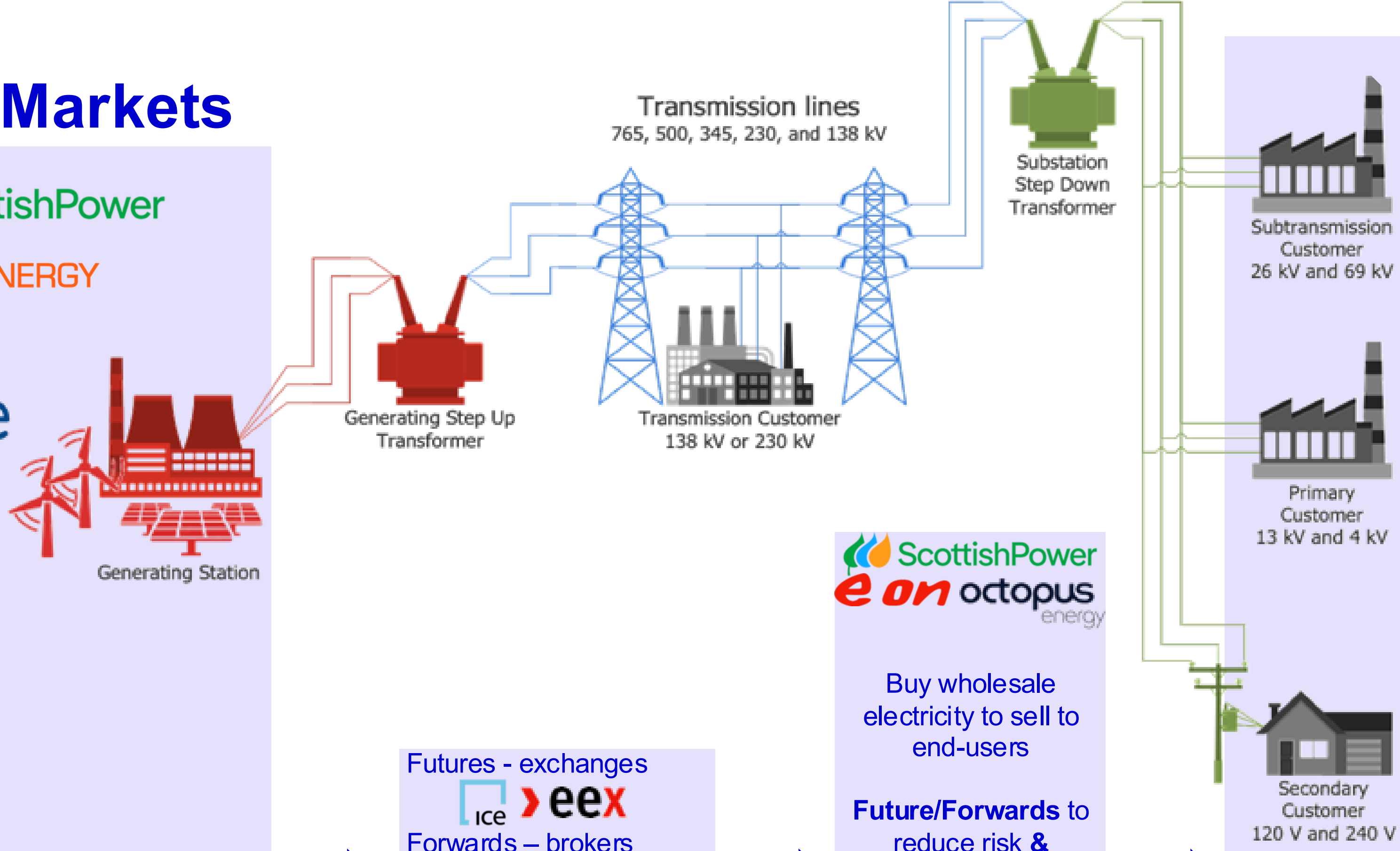
ScottishPower
e.on octopus energy

SUPPLIERS WITHOUT GENERATION

VERTICALLY INTEGRATED SUPPLIERS

GENERATORS		WHOLESALE MARKET		SUPPLIERS / RETAILERS		END-USERS
Produce electricity from various sources	Sell electricity →	Buy/Sell electricity to meet demand	Buy electricity →	Buy wholesale electricity to sell to end-users	Sell electricity →	Buy retail electricity

3. UK Markets



ScottishPower
EDF ENERGY
RWE
sse

Produce electricity from various sources

GENERATORS

Sell electricity

Futures - exchanges
ICE eex
Forwards – brokers
MAREX ICAP
Day ahead NORD POOL
Intraday epexspot

WHOLESALE MARKET

Buy electricity

ScottishPower
e.on octopus energy

Buy wholesale electricity to sell to end-users

Future/Forwards to reduce risk **&** **Spot** to "Shape" with demand

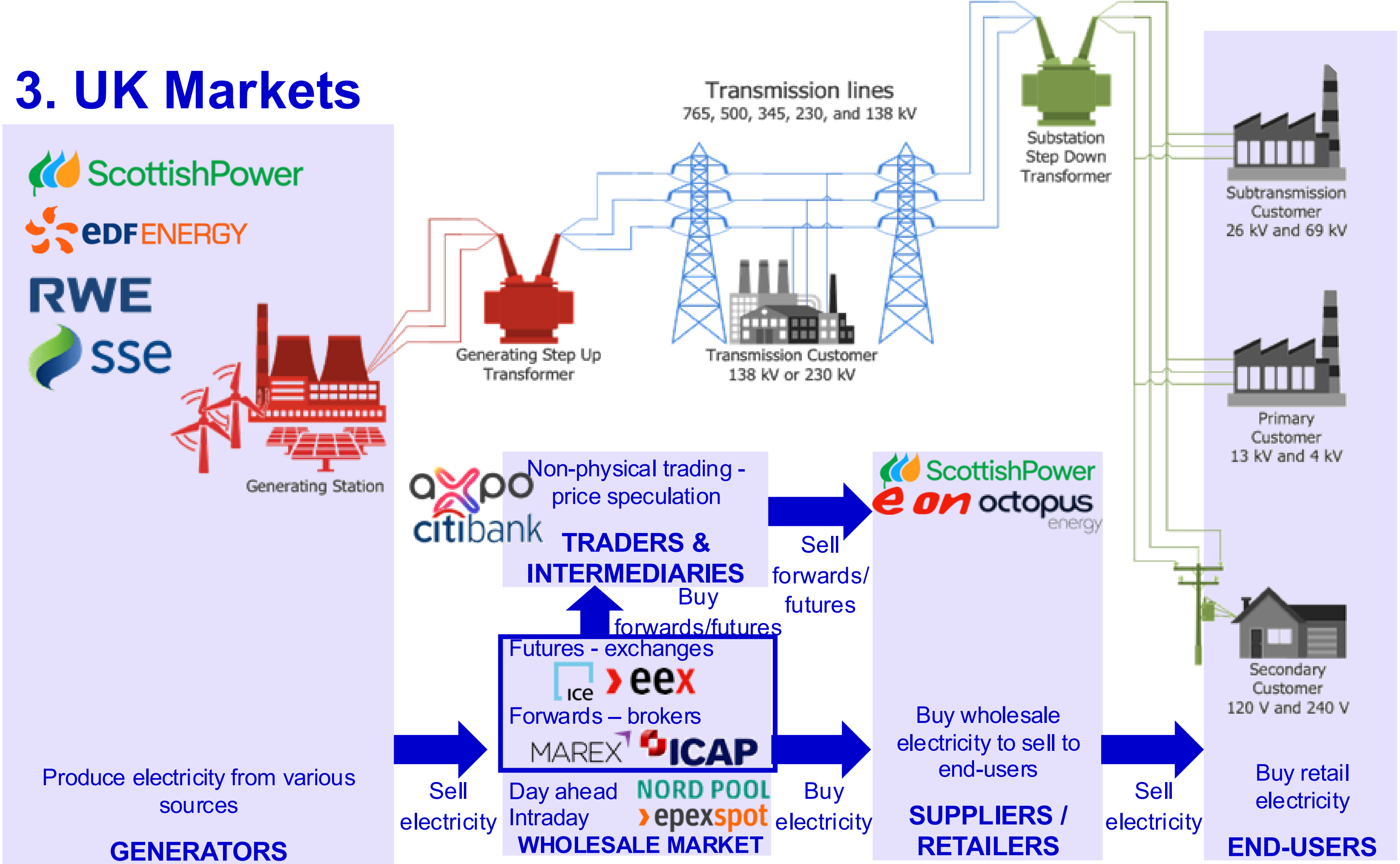
SUPPLIERS / RETAILERS

Sell electricity

Buy retail electricity

END-USERS

3. UK Markets


ScottishPower
EDF ENERGY
RWE
sse
Generating Station
Generating Step Up Transformer
Transmission lines
765, 500, 345, 230, and 138 kV
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138 kV or 230 kV
Substation Step Down Transformer
Subtransmission Customer
26 kV and 69 kV
Primary Customer
13 kV and 4 kV
Secondary Customer
120 V and 240 V
axpo
citibank
Non-physical trading - price speculation
TRADERS & INTERMEDIARIES
Buy forwards/futures
Sell forwards/ futures
ScottishPower
e on
octopus energy
Futures - exchanges
ICE
eex
Forwards – brokers
MAREX
ICAP
Day ahead
Intraday
NORD POOL
epexspot
WHOLESALE MARKET
Produce electricity from various sources
GENERATORS
Sell electricity
Buy electricity
Buy wholesale electricity to sell to end-users
SUPPLIERS / RETAILERS
Sell electricity
Buy retail electricity
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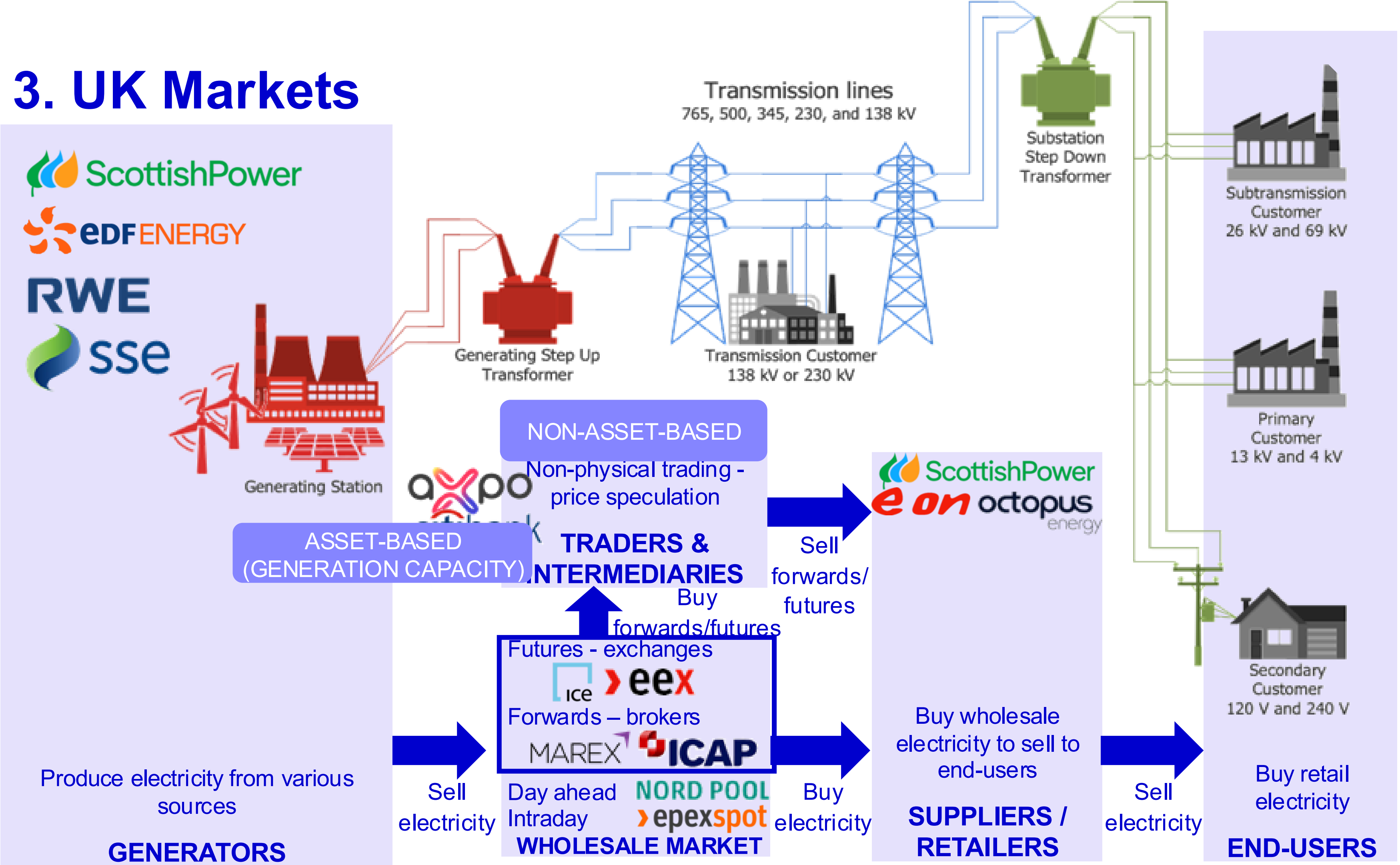
3. UK Markets


ScottishPower
EDF ENERGY
RWE
sse
Generating Station
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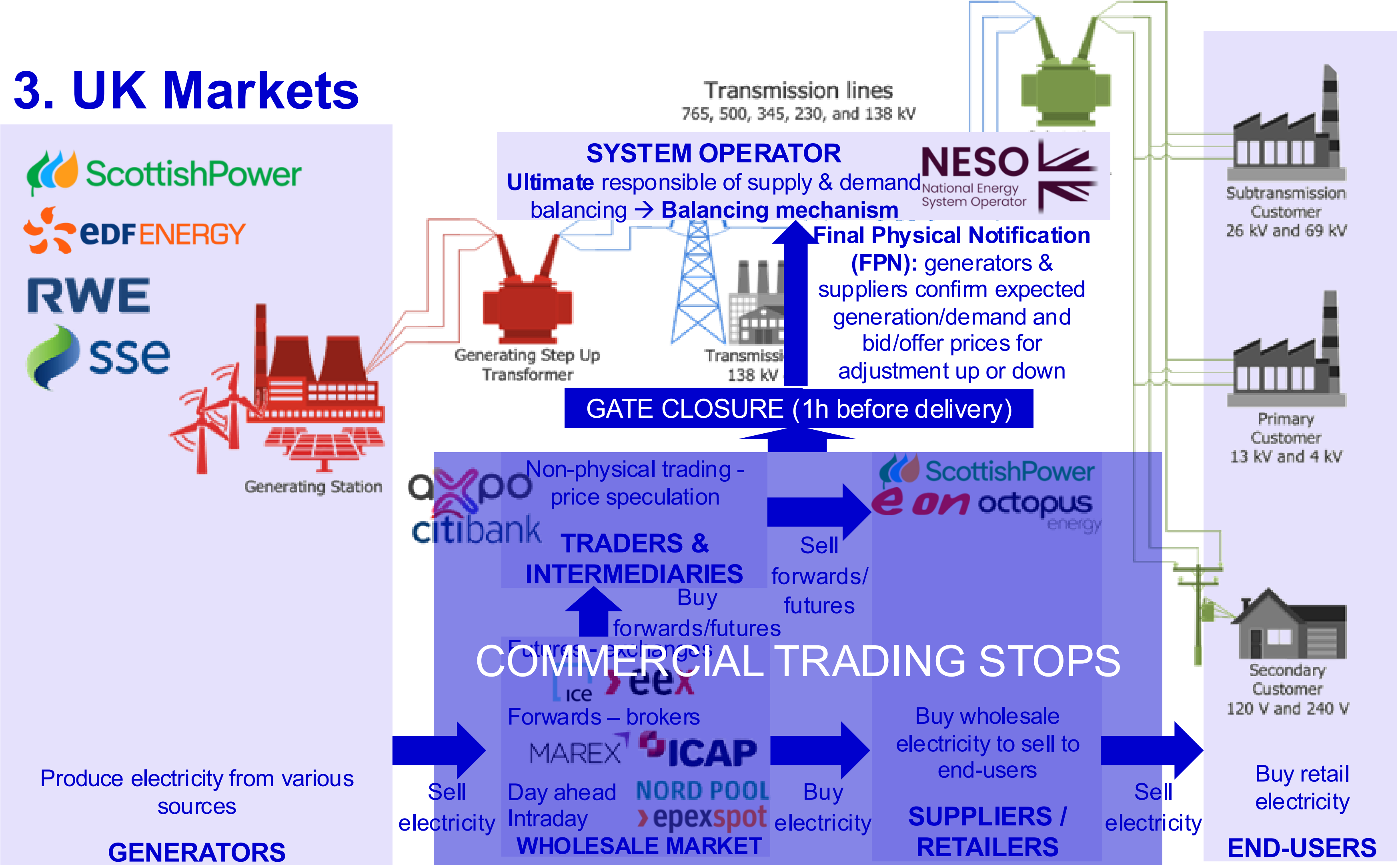
138 kV or 230 kV
Substation Step Down Transformer
Subtransmission Customer
26 kV and 69 kV
Primary Customer
13 kV and 4 kV
Secondary Customer
120 V and 240 V
NON-ASSET-BASED
Non-physical trading - price speculation
axpo
ASSET-BASED (GENERATION CAPACITY)
TRADERS & INTERMEDIARIES
Sell forwards/ futures
Buy forwards/futures
Futures - exchanges
ICE
eex
Forwards – brokers
MAREX
ICAP
Day ahead
Intraday
NORD POOL
epexspot
WHOLESALE MARKET
ScottishPower
e on
octopus energy
Buy wholesale electricity to sell to end-users
SUPPLIERS / RETAILERS
Produce electricity from various sources
GENERATORS
Sell electricity
Buy electricity
Sell electricity
Buy retail electricity
END-USERS

3. UK Markets



Transmission lines
765, 500, 345, 230, and 138 kV
SYSTEM OPERATOR
Ultimate responsible of supply & demand balancing → **Balancing mechanism**
NESO
National Energy System Operator
ScottishPower
EDF ENERGY
RWE
sse
Generating Station
Generating Step Up Transformer
Transmissi
138 kV
Final Physical Notification (FPN): generators & suppliers confirm expected generation/demand and bid/offer prices for adjustment up or down
GATE CLOSURE (1h before delivery)
Subtransmission Customer
26 kV and 69 kV
Primary Customer
13 kV and 4 kV
Secondary Customer
120 V and 240 V
axpo
citibank
Non-physical trading - price speculation
TRADERS & INTERMEDIARIES
Sell forwards/ futures
ScottishPower
e on
octopus energy
Buy forwards/futures
Futures - exchanges
COMMERCIAL TRADING STOPS
ice
eex
Forwards – brokers
MAREX
ICAP
Day ahead
Intraday
NORD POOL
epexspot
WHOLESALE MARKET
Produce electricity from various sources
GENERATORS
Sell electricity
Buy electricity
Buy wholesale electricity to sell to end-users
SUPPLIERS / RETAILERS
Sell electricity
Buy retail electricity
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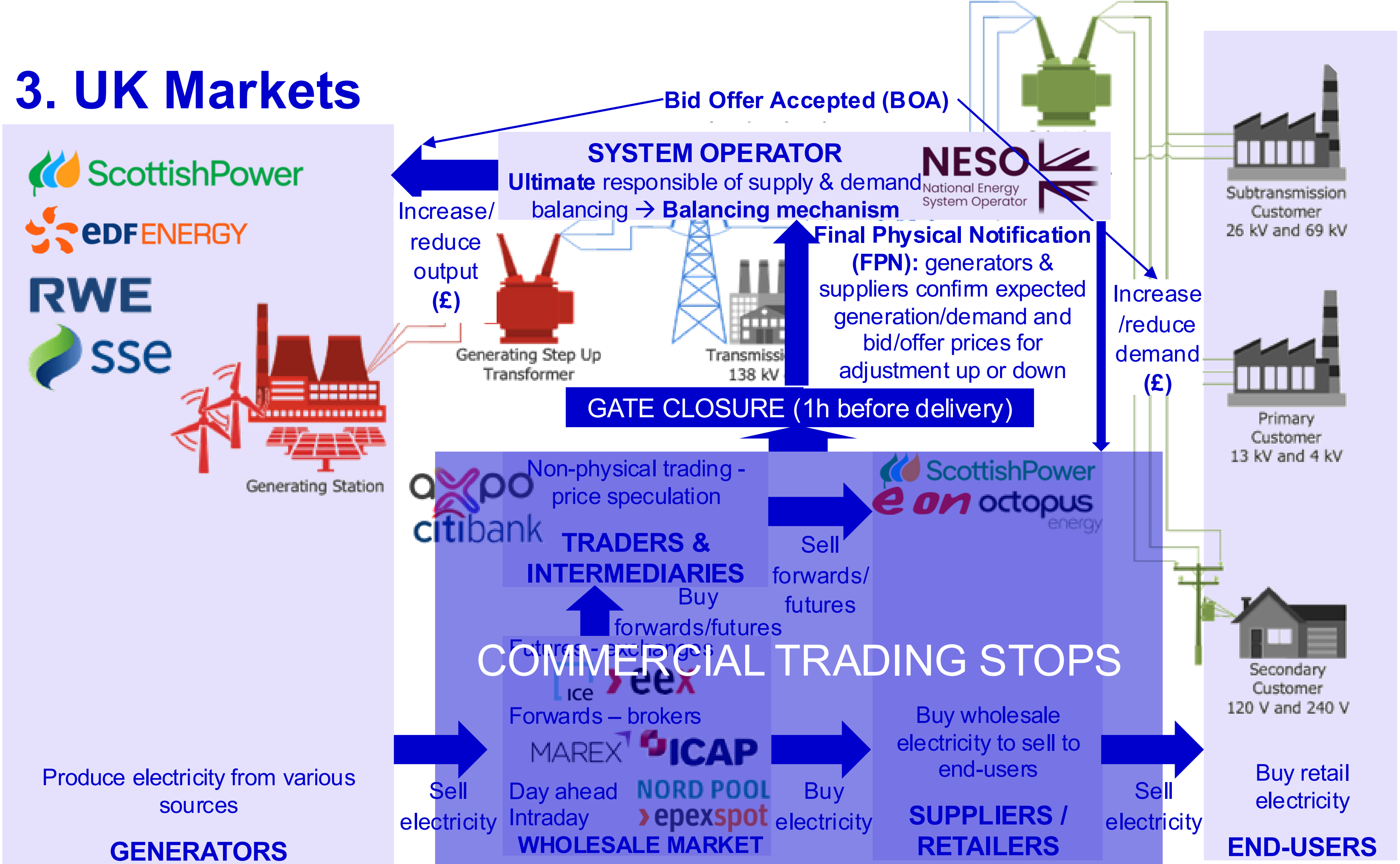

Bid Offer Accepted (BOA)
SYSTEM OPERATOR
Ultimate responsible of supply & demand balancing → Balancing mechanism
NESO
National Energy System Operator
Increase/ reduce output (£)
Generating Step Up Transformer
Transmissi
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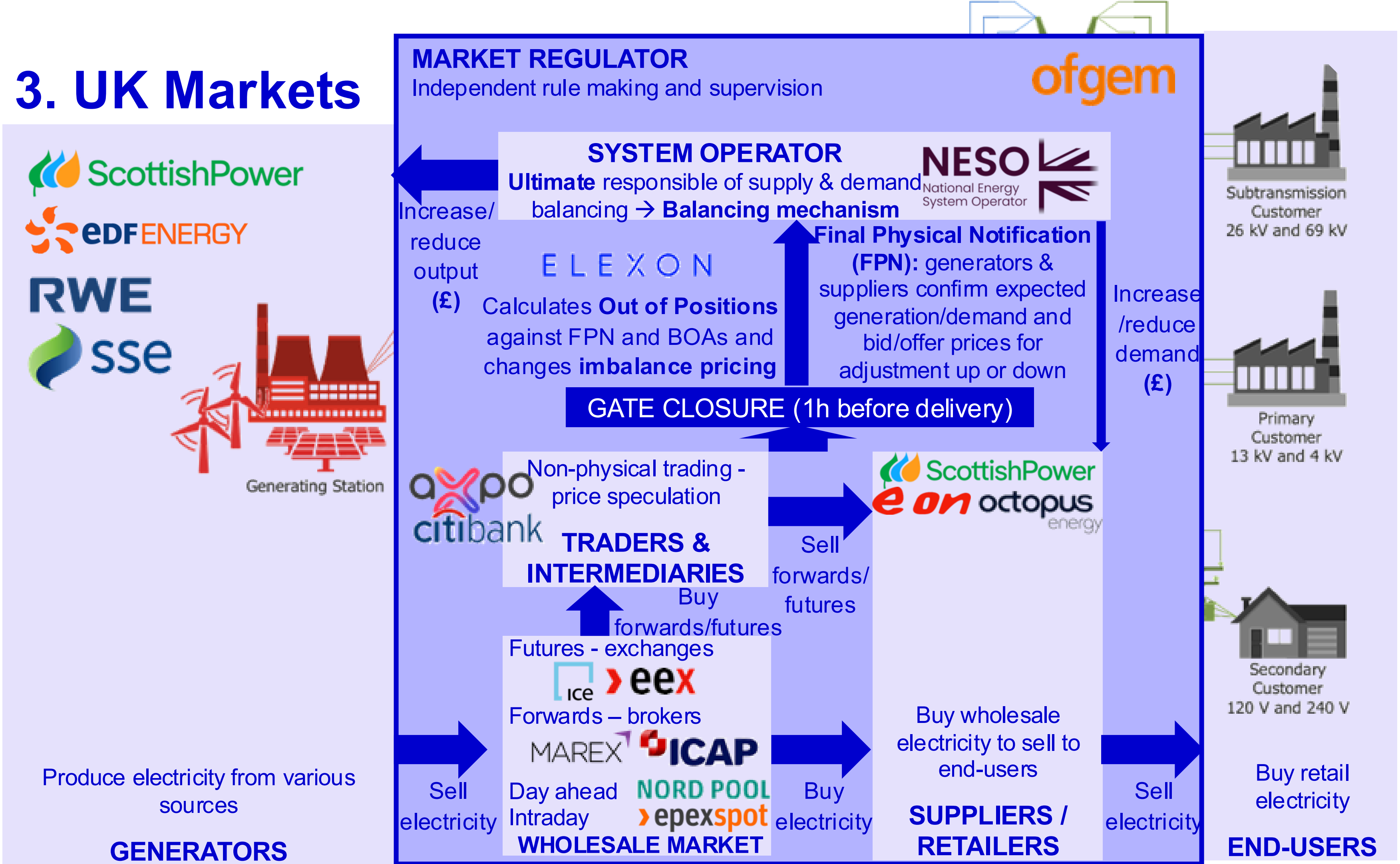

MARKET REGULATOR
Independent rule making and supervision
ofgem
SYSTEM OPERATOR
Ultimate responsible of supply & demand balancing → Balancing mechanism
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National Energy System Operator
Increase/ reduce output (£)
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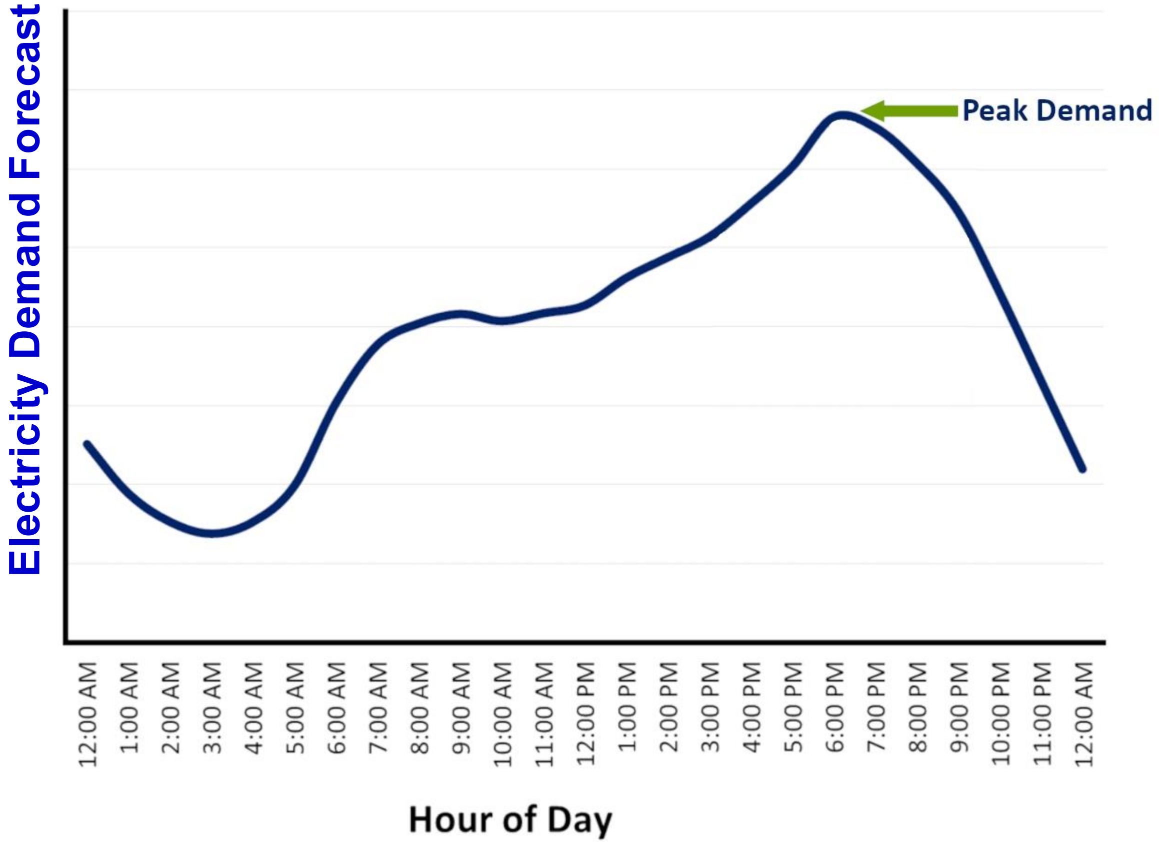
Agenda

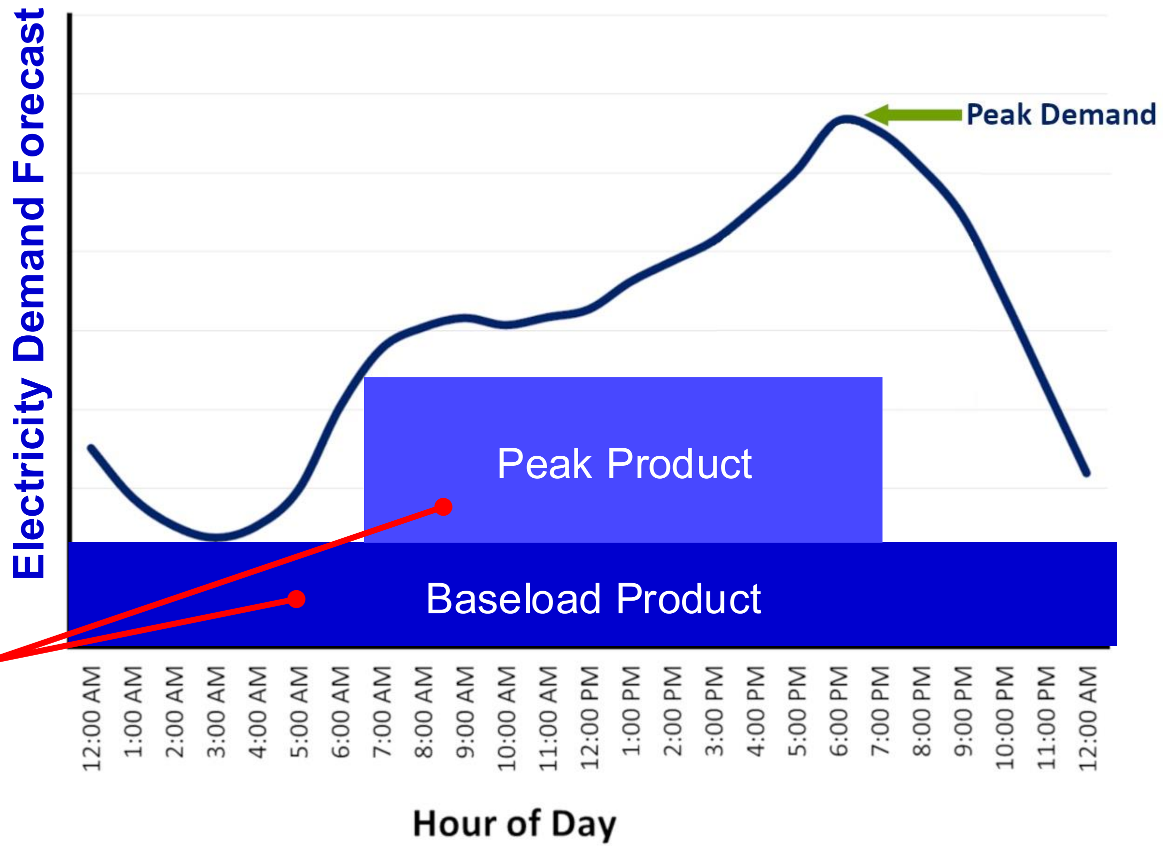
4. Electricity Pricing & Analytics

Demand follows strong patterns → plan purchasing strategy.



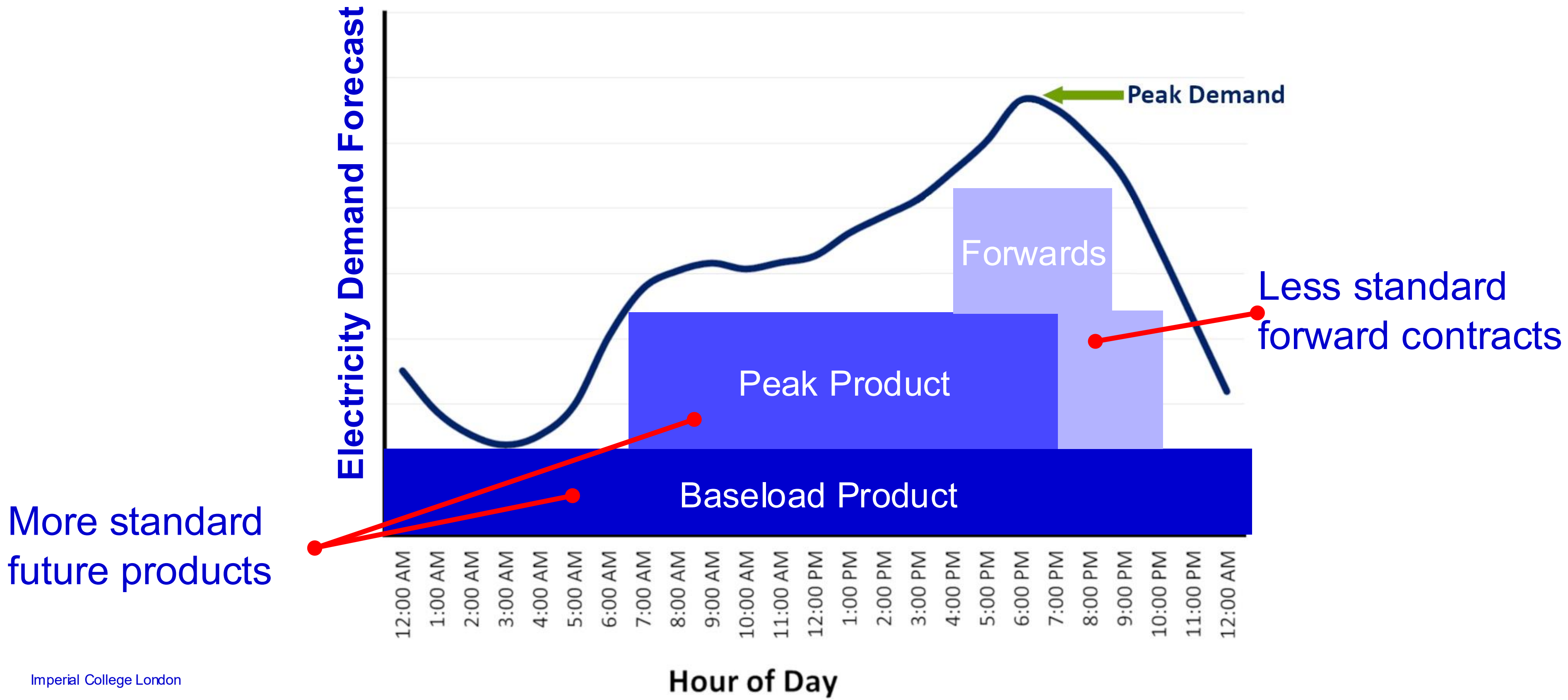
4. Electricity Pricing & Analytics

More standard future products



Reference: Modo Energy - Electricity Markets in Great Britain - The Energy Academy (Series Three)

4. Electricity Pricing & Analytics



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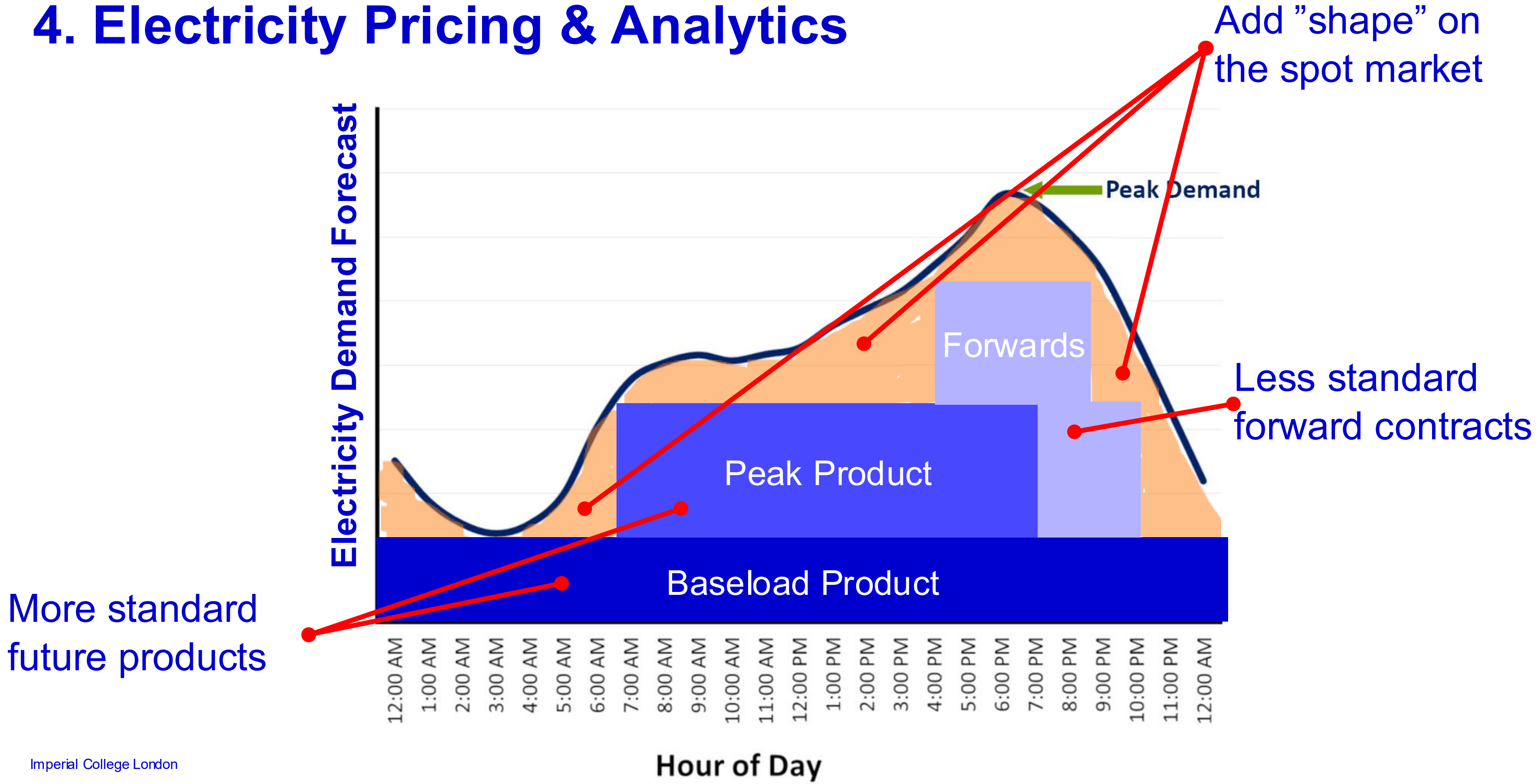

Reference: Modo Energy - Electricity Markets in Great Britain - The Energy Academy (Series Three)

4. Electricity Pricing & Analytics

30 min

GENERATORS
Produce electricity for various sources

REQUEST KWH

SUPPLIERS / RETAILERS
Buy wholesale electricity to sell to end-users

WHOLESALE MARKET
Buy/Sell electricity to meet demand

4. Electricity Pricing & Analytics

GENERATORS
Produce electricity for various sources

BID

Generators submit their volumes and price, and they are ranked by price – free market rules apply

WHOLESALE MARKET
Buy/Sell electricity to meet demand

SUPPLIERS / RETAILERS
Buy wholesale electricity to sell to end-users

4. Electricity Pricing & Analytics

GENERATORS
Produce electricity for various sources

BID

free market rules reminder

Generators produce identical products (kWh) with different production cost

Generator 1: 1 kWh
Production cost **2£**

Generator 2: 1 kWh
Production cost **6£**

Generator 3: 1 kWh
Production cost **15£**

If the market wants 1 kWh, Generator 1 can price at 6£, because:

- Generator 2 can't sell for less than 6£ without incurring loses → lower price would be a miss opportunity to Generator 1.
- Asking for a higher price could imply losing the sale to Generator 2.

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Buy/Sell electricity to meet demand

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4. Electricity Pricing & Analytics

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free market rules reminder

Generators produce identical products (kWh) with different production cost

Generator 1: 1 kWh
Production cost **2£**

Generator 2: 1 kWh
Production cost **6£**

Generator 3: 1 kWh
Production cost **15£**

If the market wants 2 kWh, Generator 1 and Generator 2 can price at 15£, because:

- Generator 3 can't sell for less than 15£ without incurring loses → lower price would be a miss opportunity to Generator 1 and Generator 2.
- Asking for a higher price could imply losing the sale to Generator 3.

WHOLESALE MARKET
Buy/Sell electricity to meet demand

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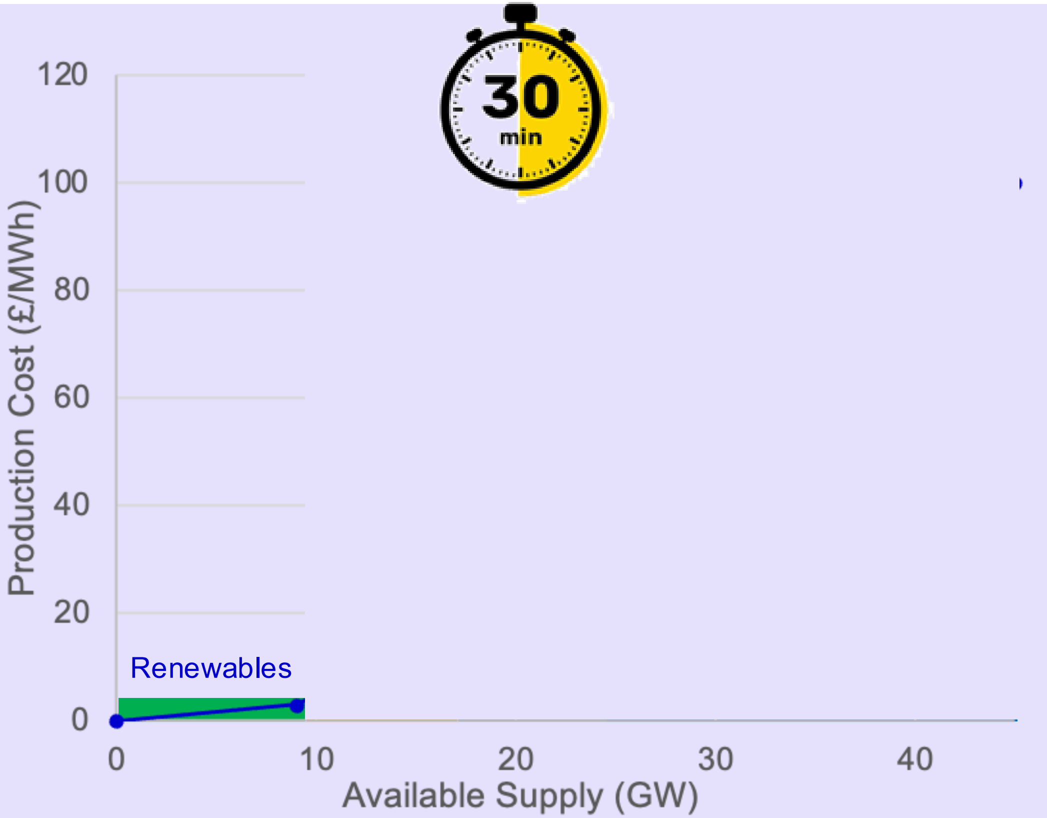
BID



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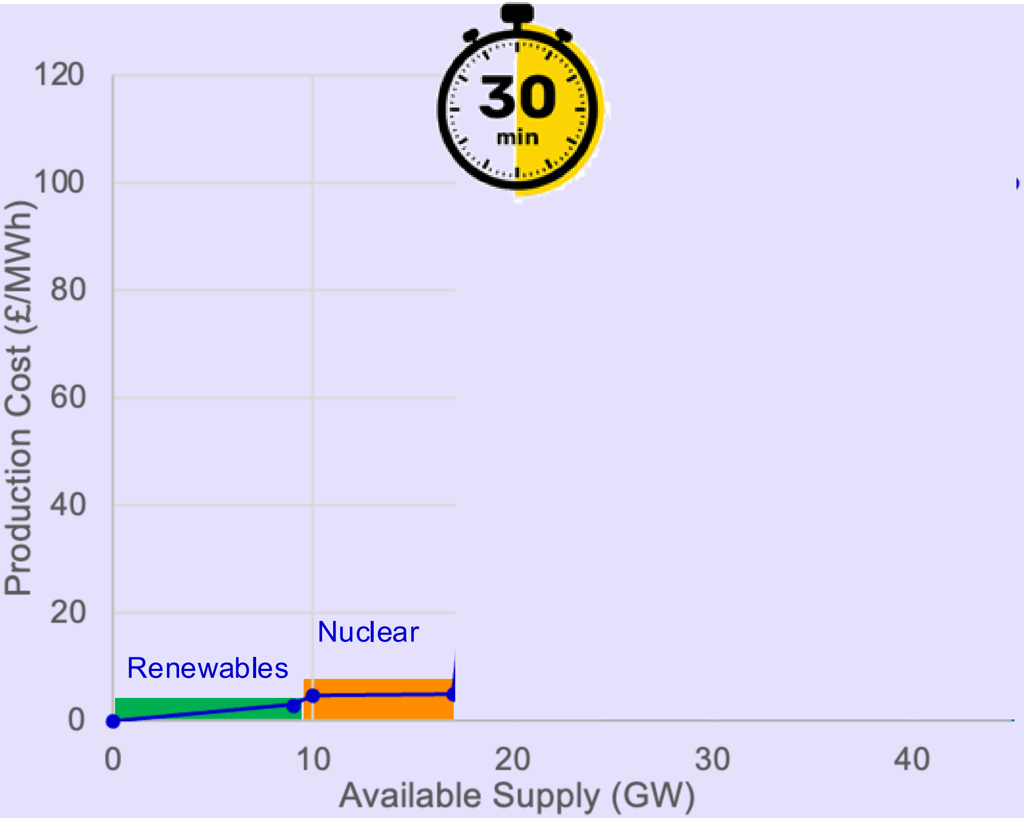

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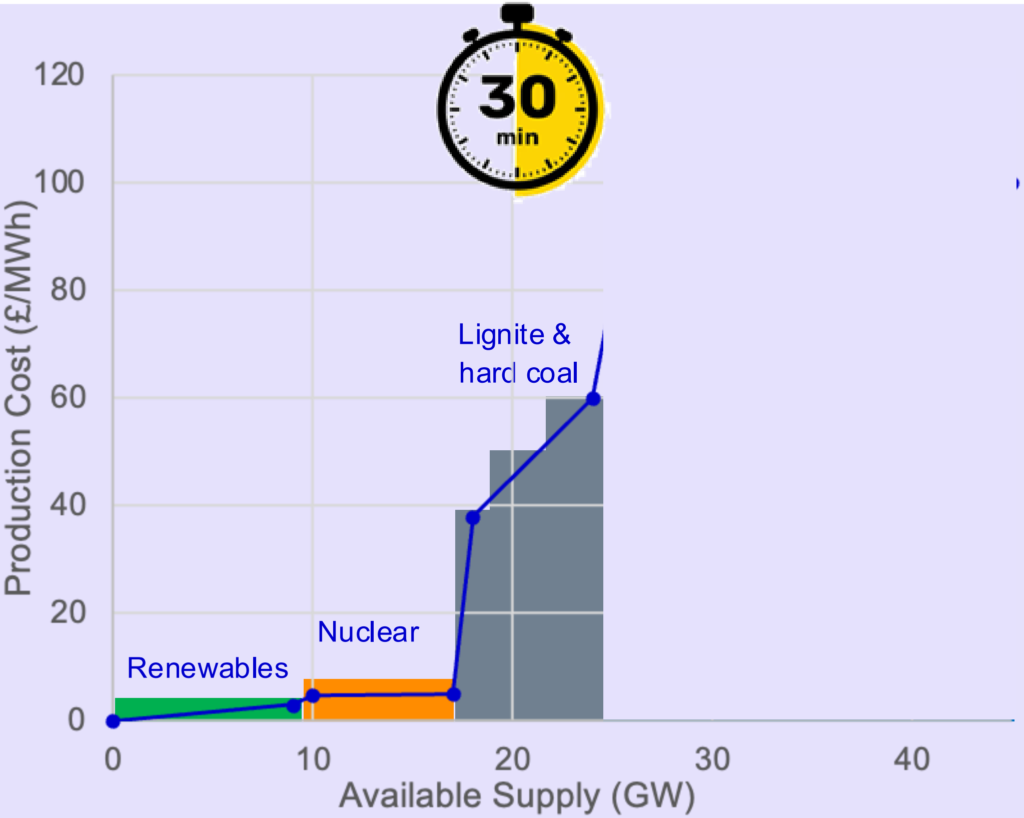

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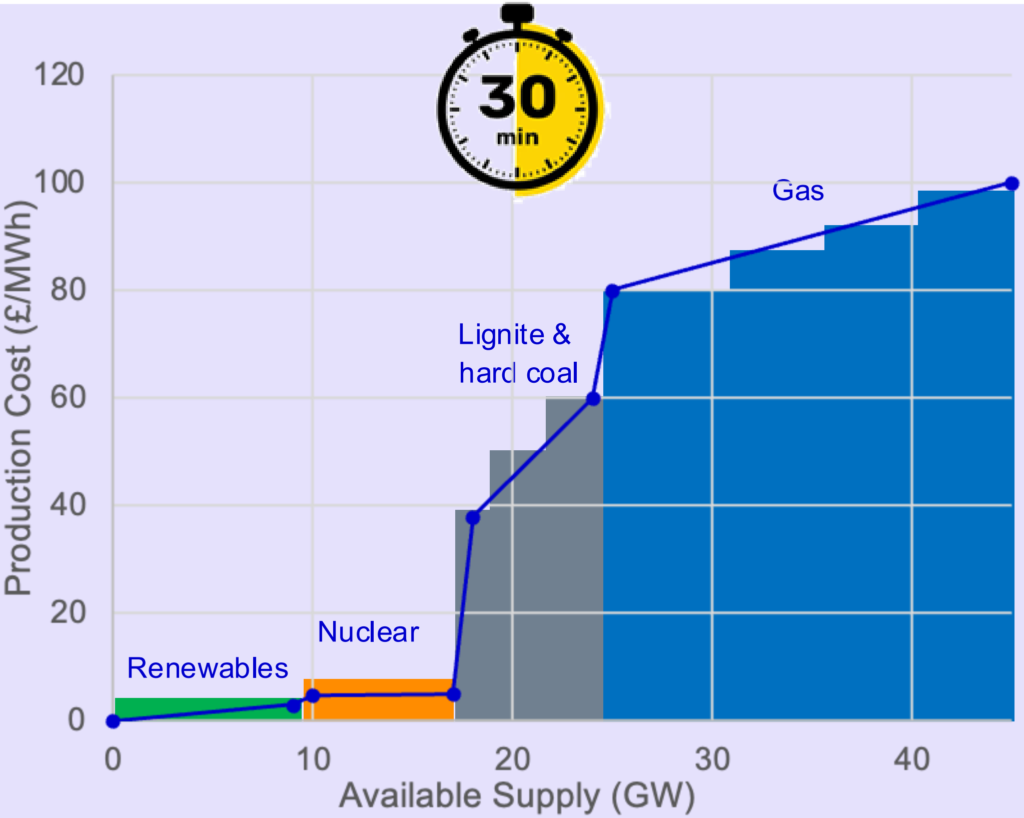

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ScottishPower
EDF ENERGY
RWE sse

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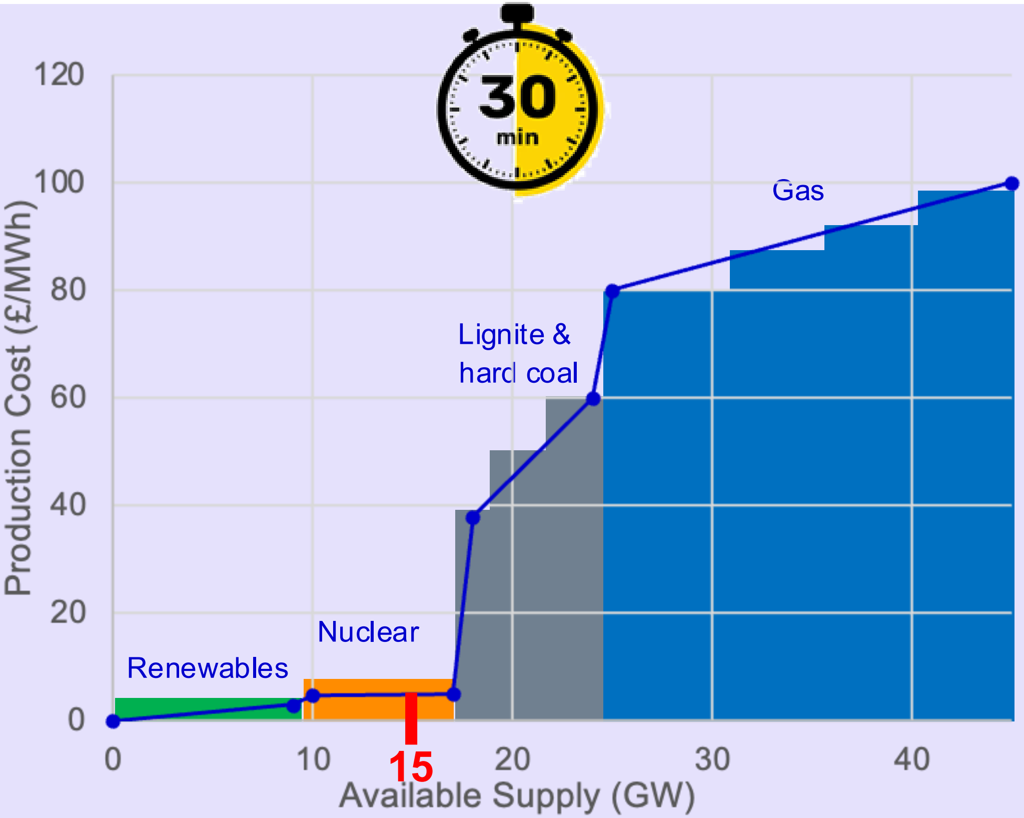

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ScottishPower
e.on octopus energy

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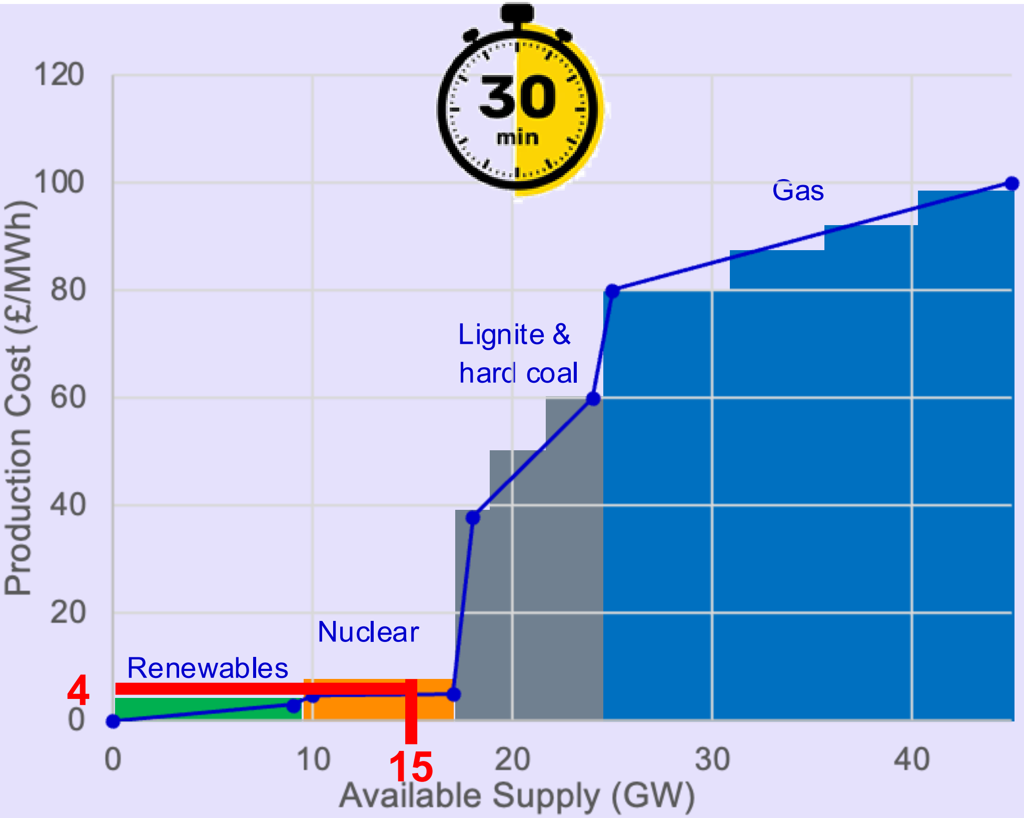

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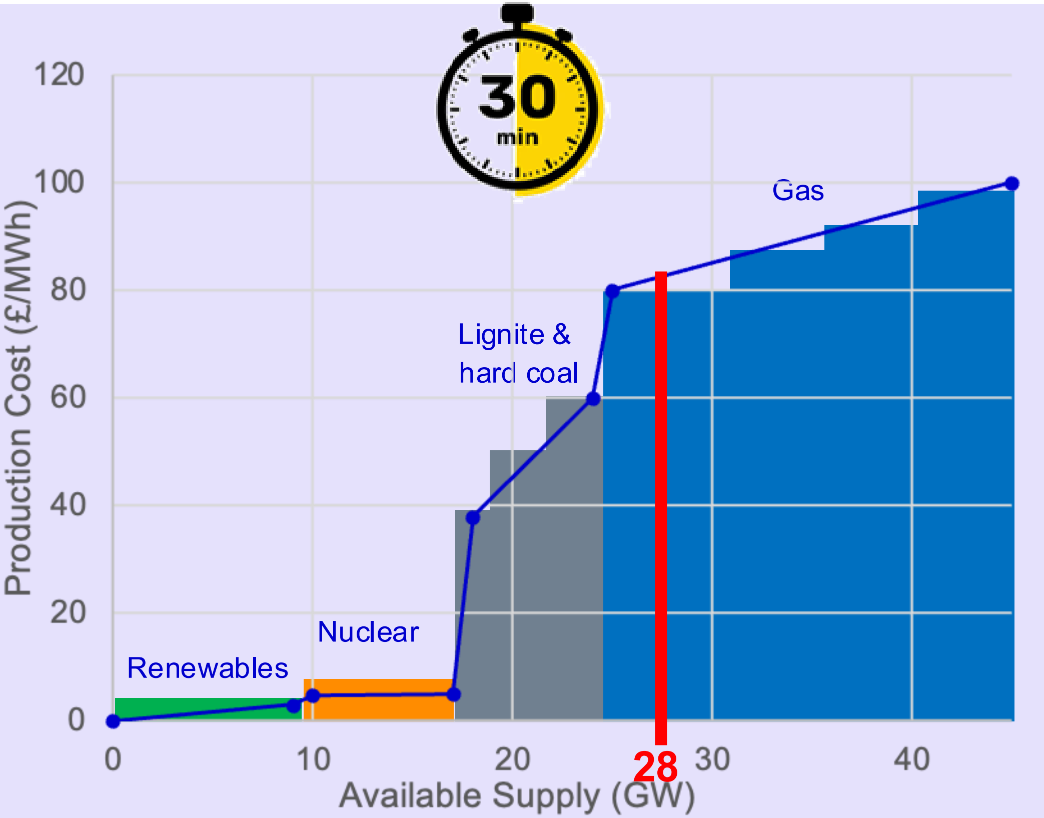

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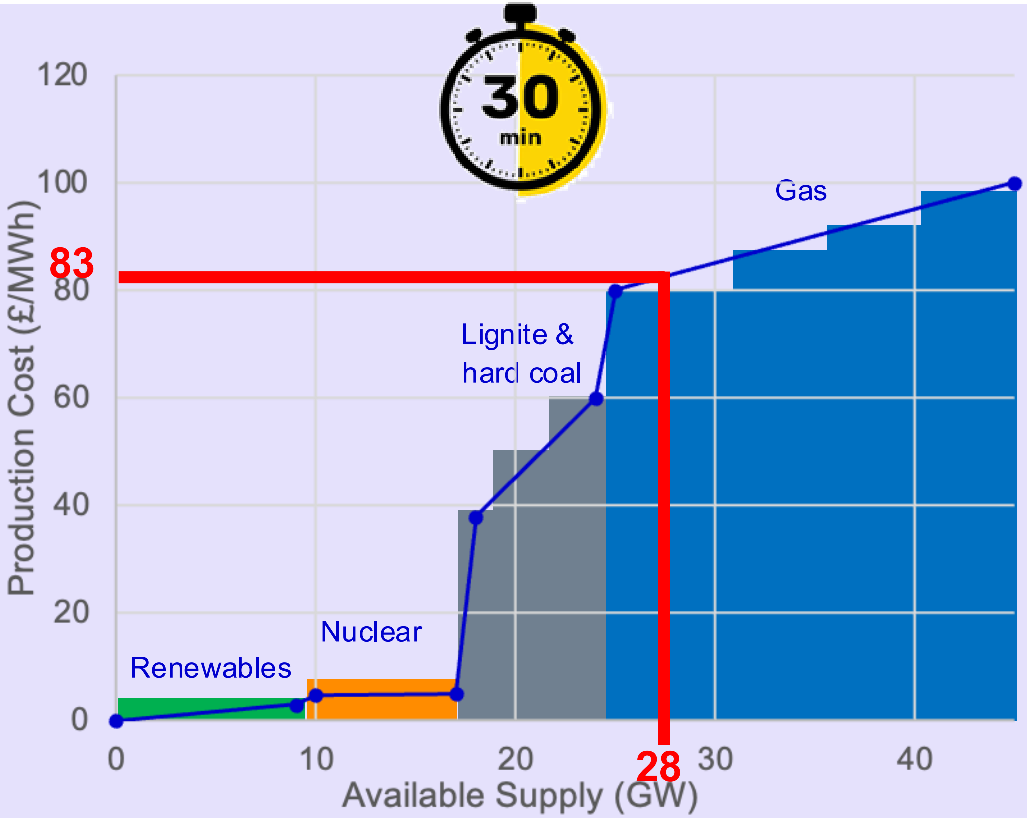

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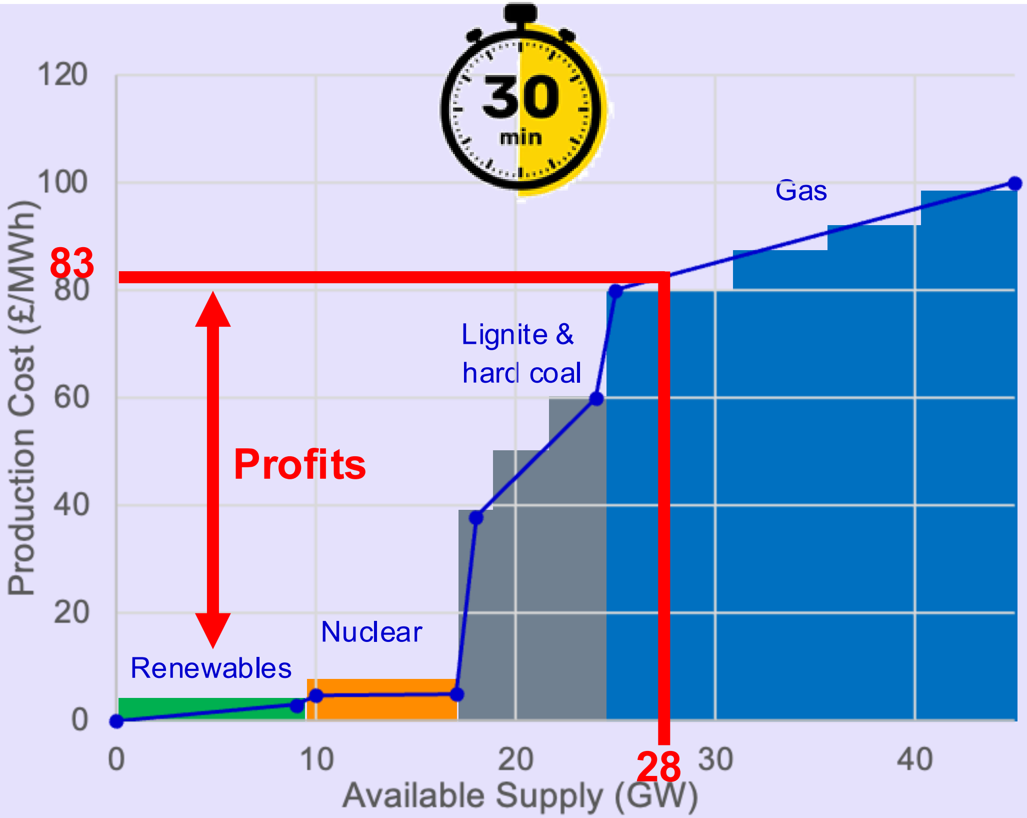

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Agenda

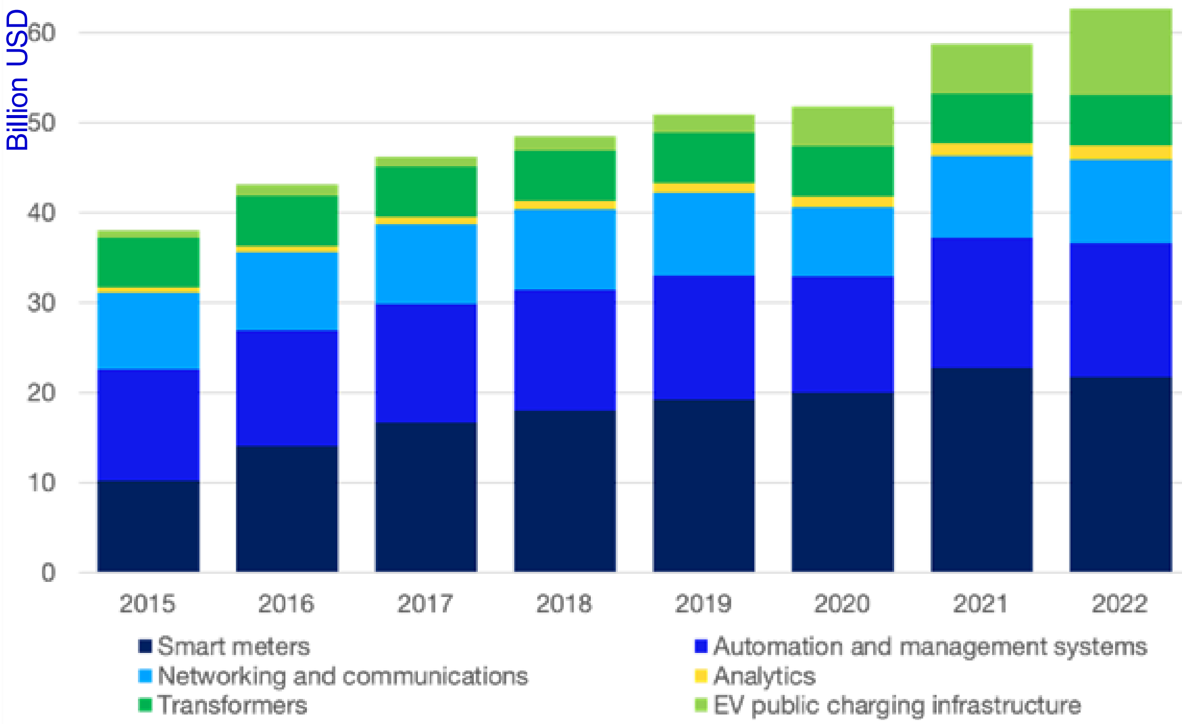
5. Electricity Analytics

Digital grid spending **20% (2022) of total grid spending.**

Applications of digitalization:

- **Operations:** Real-time monitoring, smarter load balancing.
- **Maintenance:** Predictive analytics reduces downtime and extends asset life.
- **Consumers:** Smart meters enable active participation and demand response.
- **Resilience:** Automated fault detection and faster recovery.

Digital-related investment in grid technologies, 2015-2022



Ref: https://www.iea.org/data-and-statistics/charts/digital-related-investment-in-grid-technologies-2015-2022

6. Electricity Analytics

- **Demand forecasting:** time series models (SARIMA, Prophet, LSTM) at 30-min granularity
- **Renewable generation forecasting:** weather models + ML to estimate wind & solar output and minimise imbalance costs
- **Price forecasting:** fuel prices, renewable output with tail-risk models
- **Generation scheduling:** optimisation selects which plants run at what output based on cost, availability, and constraints
- **Intraday & real-time monitoring:** continuous reforecasting as weather and demand data update
- **Portfolio optimisation:** bid optimisation across spot, futures, and forwards
- **Post-event analysis:** ML and statistical tools to analyse system actions and improve future forecast accuracy

Source: (2020) Eric O'Shaughnessy, Jesse R. Cruce, Kaifeng Xu,

Agenda

1 **Grid Architecture**

2 **Electricity Market Fundamentals**

3 **Electricity Market Models & the UK Markets**

4 **Electricity Pricing**

5 **Electricity Analytics**

6 **Data Explorer**

Data Explorer

In groups, **choose one country, explore and discuss** insights about the electricity industry, such as electricity demand, generation composition, carbon emissions, and dependencies with other countries. Share with the class something that you have learned about the country that has surprised you!

Real-Time Electricity Tracker – Data Tools - IEA

https://bit.ly/Electricity-Tracker

Balancing the Grid - Game

In this interactive game, you will dive into a day of grid balancing to understand the importance of digitalization and data reliability in managing the process.

https://www.neso.energy/energy-101/balancing-grid-interactive-game

https://bit.ly/PowerGridDemo

IMPERIAL

Thank you

Cristina Vera www.linkedin.com/in/c-rv